MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

Statement (I) : A uniform electric field and a uniform magnetic field are pointed in the same direction. If an electron is projected in the same direction the electron velocity will decrease in magnitude.
Statement (II) : Two infinite long parallel wires are carrying equal current in the same direction. The magnetic field at a point midway between the wires is zero.
Statement (III) : No net force acts on a rectangular coil carrying a steady current when suspended in a uniform magnetic field.
Which of the following is correct?

(A) Statements I, II and III are true.

(B) Statements I and II are true, but statement III is false.

(C) Statements II and III are true, but statement I is false.

(D) Statements I and III are true, but statement II is false.

Q.2

The wavelength of the first spectral line of sodium is 5896 Å. The first excitation potential of sodium atom will be ($h = 6.63 \times 10^{-34}$ J s)

(A) 4.2 V

(B) 3.5 V

(C) 2.1 V

(D) None of these

Q.3

The pressure and density of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes adiabatically from (P, d) to (P', d'). If $\frac{d'}{d} = 32$. Then, $\frac{P'}{P}$ is

(A) $\frac{1}{128}$

(B) 32

(C) 128

(D) 256

Q.4

Statement (A): When the temperature increases the viscosity of gases increases and the viscosity of liquids decreases.
Statement (B): Water does not wet an oily glass because cohesive force of oil is less than that of water.
Statement (C) : A liquid will wet a surface of a solid if the angle of contact is greater than 90°.

(A) A. B. and C are false

(B) A and B false, C is true

(C) B and C false, A is true

(D) A and C false, B is true

Q.5

In a slide caliper, $(m+1)$ number of vernier divisions is equal to m number of smallest main scale divisions. If d unit is the magnitude of the smallest main scale division, then the magnitude of the vernicr constant is

(A) $d/(m+1)$ unit

(B) d/m unit

(C) md/ $(m+1)$ unit

(D) $(m+1)d/m$ unit

Q.6

A particle is released freely from a height H. At a certain height, its kinetic energy is two times of its potential energy. Then, the height and the speed of the particle at that instant are respectively
($g=$ acceleration due to gravity)

(A) $\frac{H}{3}, \sqrt{\frac{2gH}{3}}$

(B) $\frac{H}{3}, 2\sqrt{\frac{gH}{3}}$

(C) $\frac{2H}{3}, \sqrt{\frac{2gH}{3}}$

(D) $\frac{H}{3}, \sqrt{2gH}$

Q.7

A copper disc of the radius $0.1\ \mathrm{m}$ is rotated about its centre with $\omega = 20$ revolutions per second in a uniform magnetic field of $0.1\ \mathrm{T}$ with its plane perpendicular to the field. The emf induced across the radius of the disc is

(A) $\frac{\pi}{20}\ \mathrm{V}$

(B) $\frac{\pi}{10}$ V

(C) $20\pi\mathrm{mV}$

(D) $100\pi\mathrm{mV}$

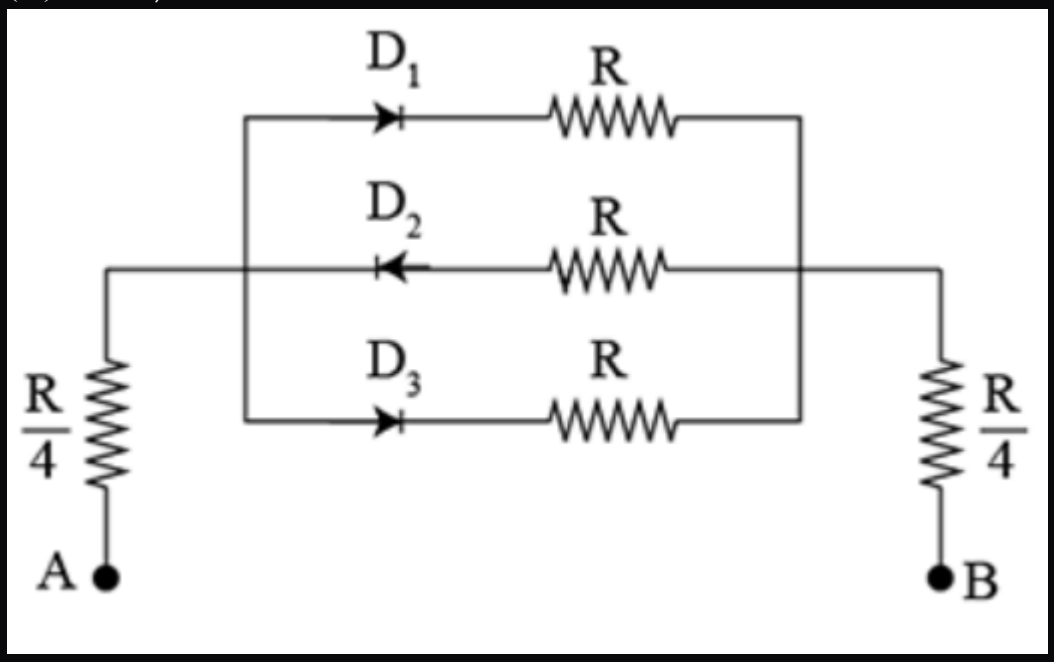
Q.8

In the following circuits, PN-junction diodes $\mathrm{D}_1, \mathrm{D}_2$ and D_3 are ideal for the following potential of A and B, the correct increasing order of resistance between A and B will be

(i) -10 V, -5 V

(ii) -5 V, -10 V

(iii) -4 V, -12 V



(A) (i) $<$ (ii) $<$ (iii)

(B) (iii) $<$ (ii) $<$ (i)

(C) (ii) $=$ (iii) $<$ (i)

(D) (i) $=$ (iii) $<$ (ii)

Q.9

The ratio of the speed of sound in a monatomic gas at $27^\circ\mathrm{C}$ and rms speed of the molecules of the same gas at a temperature of $127^\circ\mathrm{C}$ is

(A) 1 : 2

(B) $\sqrt{5} : \sqrt{12}$

(C) 3 : 4

(D) $\sqrt{13} : \sqrt{17}$

Q.10

Three thin lenses are combined by placing them in contact with each other to get more magnification in an optical

instrument. Each lens has a focal length of $3\ \text{cm}$. If the least distance of distinct vision is taken as $25\ \text{cm}$, the total magnification of the lens combination in normal adjustment is

(A) 9

(B) 25

(C) 300

(D) 3

Q.11

n identical resistance are taken in which $\frac{n}{2}$ resistors are joined in series in the left gap and the remaining $\frac{n}{2}$ resistances are joined in parallel in the right gap of a metre bridge. Balancing length in cm is

(A) $100 \cdot \frac{n^2}{n^2+4}$

(B) $100 \cdot \frac{n^2}{n^2+1}$

(C) $400 \cdot \frac{1}{n^2+4}$

(D) $400 \cdot \frac{1}{n^2+1}$

Q.12

An infinite non-conducting sheet has a surface charge density of $7 \times 10^{-7} \text{Cm}^{-2}$ on one side. The distance between equipotential surfaces whose potentials differ by $19.8\ \text{V}$, will be
(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ SI units)

(A) $2.0\ \text{mm}$

(B) $0.25\ \text{mm}$

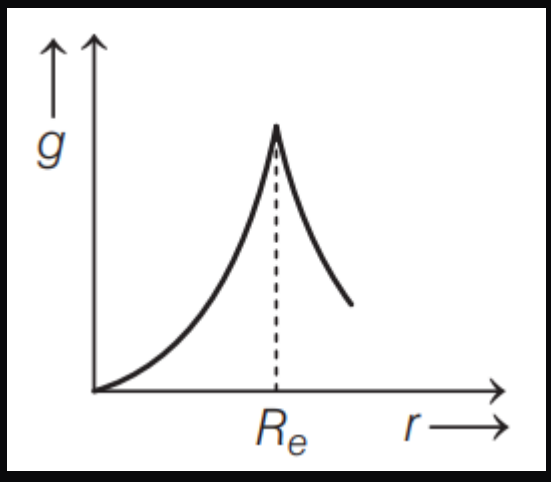
(C) $1.0\ \text{mm}$

(D) $0.5\ \text{mm}$

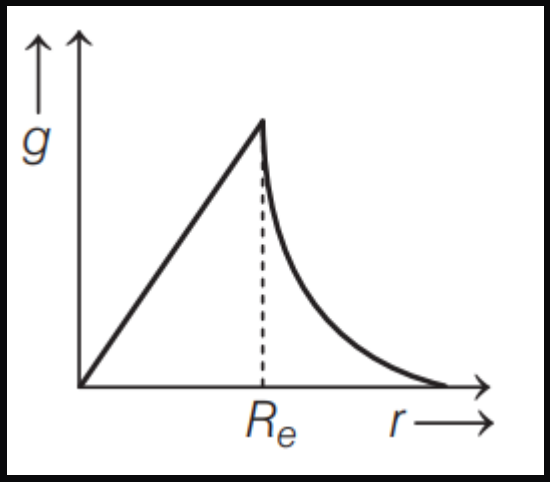
Q.13

The graph correctly represents the variation of acceleration due to gravity (g) with radial distance from the centre of the earth (radius of the earth $= R_e$) is

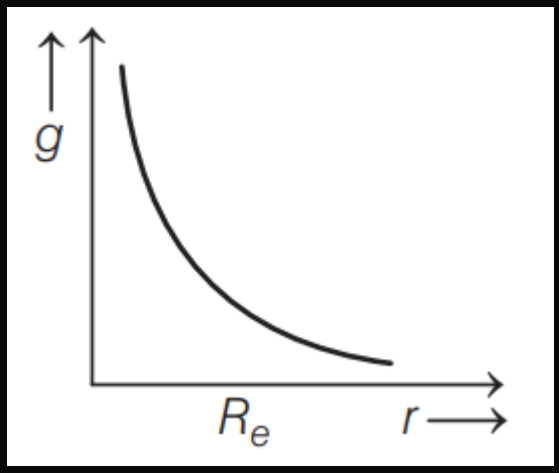
(A)



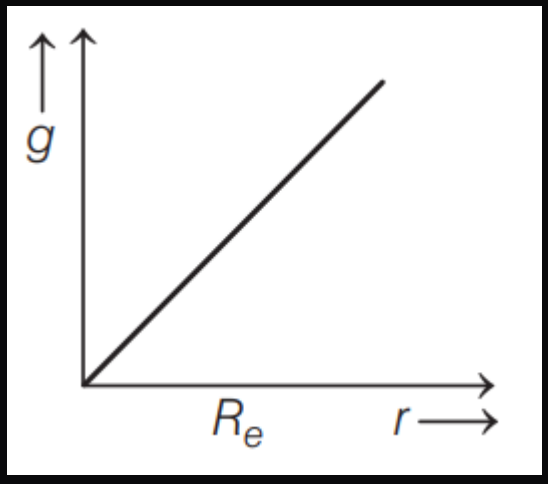
(B)



(C)



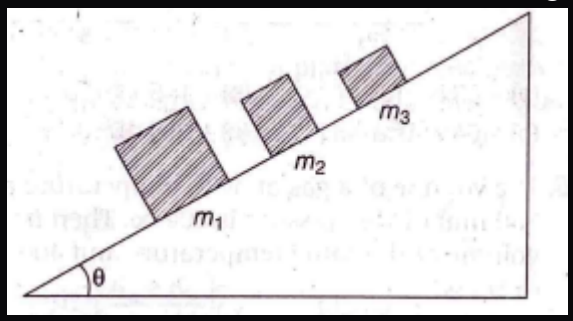
(D)



Q.14

Consider three masses m_1, m_2 and m_3 $(m_1 > m_2 > m_3)$ are at rest on a horizontal plane as shown in the figure. Now the angle of inclination (θ) of the plane is gradually increased until the masses just begin to slide. (Assume the coefficient of static friction between the masses and the surface is constant).

Then the correct statement of the following is



(A) m_3 begins to slide at a higher inclination angle than m_1 and m_2.

(B) m_3 begins to slide at a lower inclination angle than m_1 and m_2.

(C) m_1, m_2 and m_3 begins to slide at the same inclination angle.

(D) m_2 begins to slide at a higher inclination angle than m_1 and m_3.

Q.15

The speed of light (c), gravitational constant (G) and Planck's constant (h) are taken as fundamental units in a system. The dimensions of time in this new system should be:

(A) $G^{\frac{1}{2}}\ h^{\frac{1}{2}}\ c^{\frac{-5}{2}}$

(B) $G^{\frac{-1}{2}}\ h^{\frac{1}{2}}\ c^{\frac{1}{2}}$

(C) $G^{\frac{1}{2}}\ h^{\frac{1}{2}}\ c^{\frac{-3}{2}}$

(D) $G^{\frac{1}{2}}\ h^{\frac{1}{2}}\ c^{\frac{1}{2}}$

Q.16

In a plane EM wave the electric field oscillates sinusoidally at a frequency of 30MHz and amplitude $150\ \text{V/m}$. Identify the correct expression of $\overline{\text{B}}$ assuming the wave is propagating along x-axis and is oscillating along y-axis.

(A) $5 \times 10^{-7} \sin\left[\frac{x}{3} - 6 \times 10^{+7} t\right] \hat{z}\ \text{T}$

(B) $5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{5} - 6 \times 10^{+7} t\right)\right] \hat{z}\ \text{T}$

(C) $5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{10} - 3 \times 10^{+7} t\right)\right] \hat{z}\ \text{T}$

(D) $5 \times 10^{-7} \sin\left[\pi\left(\frac{2x}{5} - 6 \times 10^{+8} t\right)\right] \hat{z}\ \text{T}$

Q.17

Assertion: The resistance of a superconductor is zero.
Reason: Superconductors are used for the transmission of electric power.

(A) If both Assertion and Reason are true and Reason is correct explanation of Assertion.

(B) If both Assertion and Reason are true but Reason is not the explanation of Assertion.

(C) If Assertion is true but Reason is false.

(D) If Assertion is false but Reason is true.

Q.18

A prism is made up of material of refractive index $\sqrt{3}$. The angle of the prism is A. If the angle of minimum deviation is equal to the angle of the prism, then the value of A is:

(A) 30°

(B) 45°

(C) 60°

(D) 75°

Q.19

Match the following and find the correct pairs.

	List I		List II
(A)	Fleming's left hand rule	(i)	Direction of induced current
(B)	Right hand thumb rule	(ii)	Magnitude and direction of magnetic induction
(C)	Biot-Savart law	(iii)	Direction of force due to magnetic induction
(D)	Fleming's right hand rule	(iv)	Direction of magnetic lines due to current

(A) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)

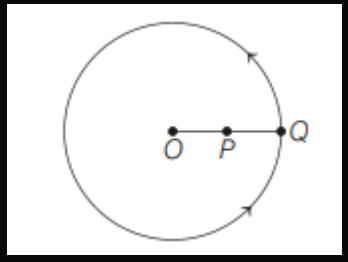
(B) (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

(C) (A)-(ii), (B)-(iv), (C)-(iii), (D)-(i)

(D) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)

Q.20

Two spheres P and Q, each of mass 200 g are attached to a string of length one metre as shown in the figure. The string and the spheres are then whirled in a horizontal circle about O at a constant angular speed. The ratio of the tension in the string between P and Q to that of between P and O is (P is at mid-point of the line joining O and Q)



(A) $\frac{1}{2}$

(B) $\frac{2}{3}$

(C) $\frac{3}{2}$

(D) $\frac{2}{1}$

Q.21

If $0.1\ \mathrm{J}$ of energy is stored for the flow of the current of $0.2\ \mathrm{A}$ in an inductor, then its inductance value (in H) is

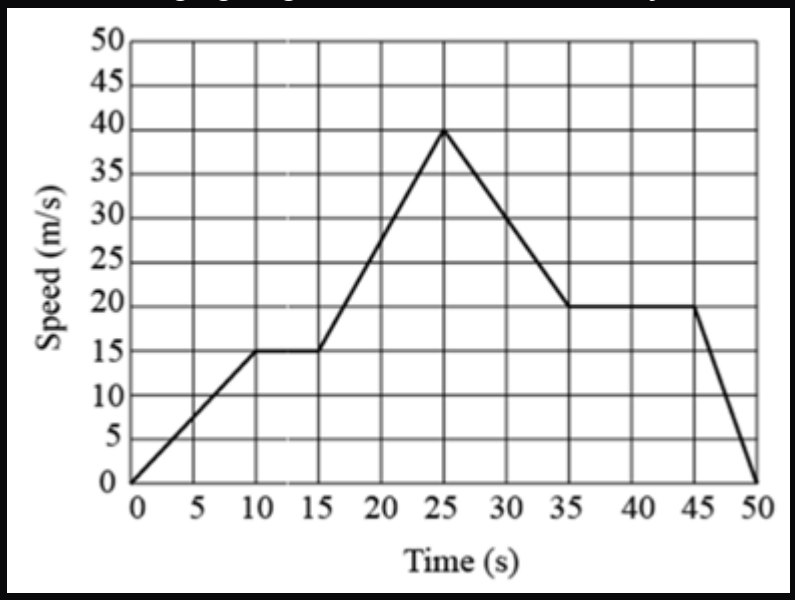
Q.22

A circular wire loop of radius $1\ \mathrm{cm}$ carries a total charge $1 \times 10^{-6}\mathrm{C}$ distributed uniformly over its length. If 0.01% of its length (circumference) is cut-off, then the electric field at the centre of the loop due to the remaining wire is ____ $\times 10^3\ \mathrm{N/C}$.

$\left(\text{Take, } \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \text{SI unit}\right)$

Q.23

The following figure gives the movement of an object.



Find total distance (in meter) travelled by object.

Q.24

Calculate the energy released in MeV in the following nuclear reaction :

$^{238}_{92}\mathrm{U} \rightarrow\ ^{234}_{90}\mathrm{Th} +\ ^{4}_{2}\mathrm{He} + \mathrm{Q}$

[Mass of $^{238}_{92}\mathrm{U}$ = 238.05079 u]

Mass of $^{234}_{90}\mathrm{Th}$ = 234.043630 u

Mass of $^{4}_{2}\mathrm{He}$ = 4.002600 u

1 u = 931.5 MeV (mark answer to nearest integer in MeV)

Q.25

A slab of side $50\ \mathrm{cm}$ and thickness $10\ \mathrm{cm}$ is subjected to a shearing force of $10^5\ \mathrm{N}$ on its narrow edge. If the

lower edge is riveted to the floor and upper edge is displaced by $0.2\ \mathrm{mm}$, then shear modulus of the material of the slab is ____ GPa.

Q.26

Which of the following statements about DNA is not correct?

(A) It has a double helix structure

(B) Adenine forms hydrogen bonds with thymine and cytosine forms hydrogen bonds with guanine

(C) The two strands in a DNA molecule are not complementary to each other

(D) It contains the pentose sugar, 2-deoxyribose

Q.27

In an atom the order of increasing energy of electrons with quantum numbers
(i) $n = 4, \ell = 1$
(ii) $n = 4, \ell = 0$
(iii) $n = 3, \ell = 2$ and (iv) $n = 3, \ell = 1$ is

(A) $\mathrm{iii} < (\mathrm{i}) < \mathrm{iv} < \mathrm{ii}$

(B) $\mathrm{ii} < \mathrm{iv} < \mathrm{i} < \mathrm{iii}$

(C) $\mathrm{i} < \mathrm{iii} < \mathrm{ii} < \mathrm{iv}$

(D) $\mathrm{iv} < \mathrm{ii} < \mathrm{iii} < \mathrm{i}$

Q.28

Assertion (A) The pH of a buffer solution containing equal moles of acetic acid and sodium acetate is 4.8 ($\mathrm{p}K_a$ of acetic acid is 4.8).
Reason (R) The ionic product of water at $25^{\circ}\mathrm{C}$ is $10^{-14}\ \mathrm{mol}^2.\mathrm{L}^{-2}$. The correct answer is

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(C) (A) is true but (R) is not true

(D) (A) is not true but (R) is true

Q.29

Assertion : Tertiary alcohols produce turbidity immediately with Lucas reagent.
Reason : Lucas reagent is a $1:1$ mixture of conc. $\mathrm{HNO_3}$ and anhydrous $\mathrm{ZnCl_2}$.
The correct option among the following is

(A) (A) is true, (R) is true and (R) is the correct explanation for (A)

(B) (A) is true, (R) is true but (R) is not the correct explanation for (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q.30

In +2 oxidation state, which of the following lanthanoids act as reducing agents?

(A) $\mathrm{Ce, Pr}$

(B) $\mathrm{Eu, Gd}$

(C) $\mathrm{Eu, Yb}$

(D) $\mathrm{Lu, Er}$

Q.31

The weight in grams of a non-volatile solute (mol. wt. 60) to be dissolved in 90 g of water to produce a relative lowering of vapour pressure of 0.02 is

(A) 4

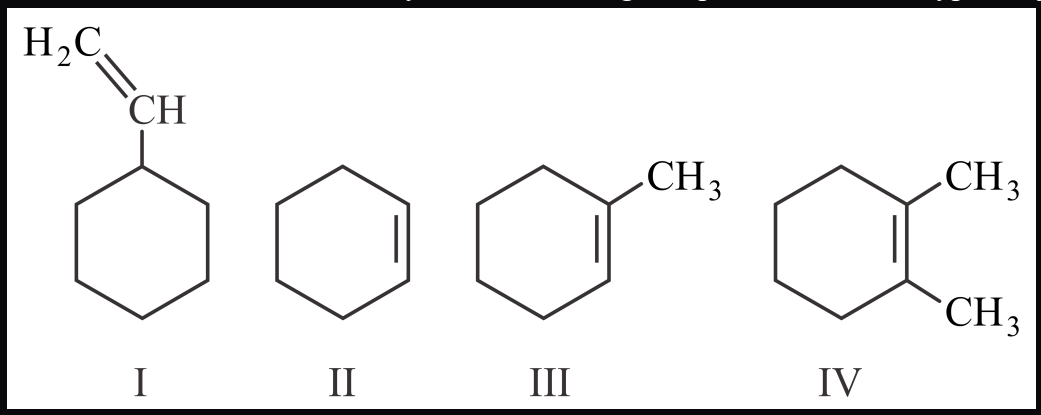
(B) 8

(C) 6

(D) 10

Q.32

The correct order of the stability of the following compounds based on hyperconjugation is



(A) $\mathrm{IV > III > II > I}$

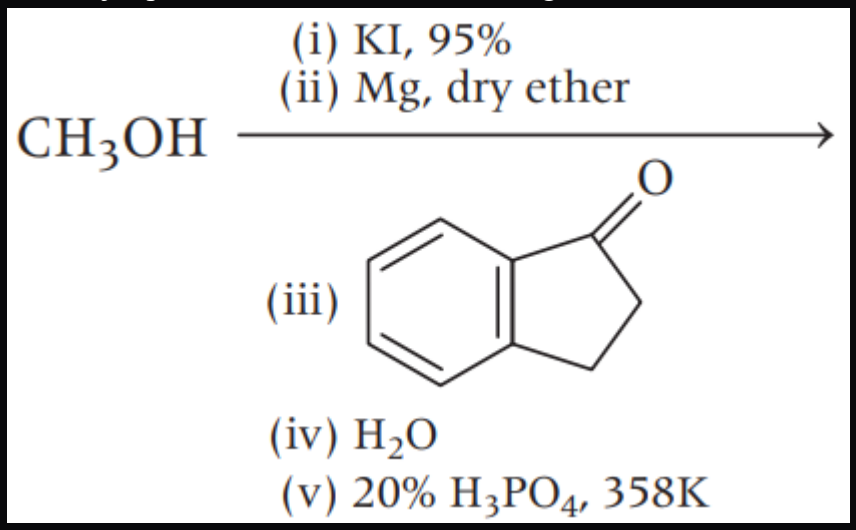
(B) $\mathrm{IV > II > I > III}$

(C) $\mathrm{IV > II > III > I}$

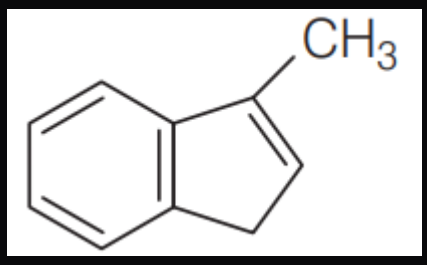
(D) $\mathrm{IV > I > III > II}$

Q.33

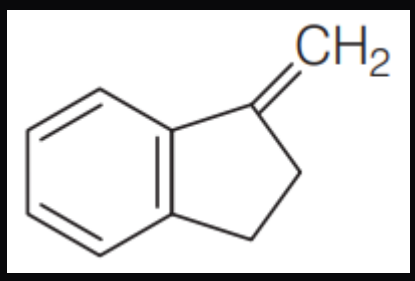
The major product formed in the following reactions is



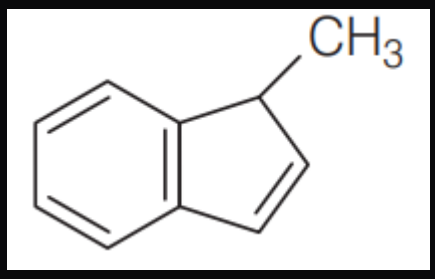
(A)



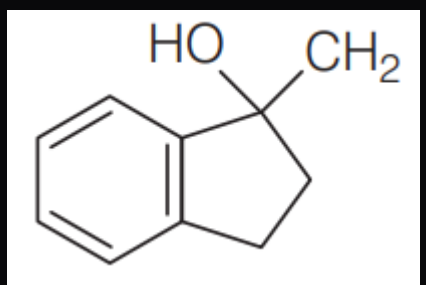
(B)



(C)



(D)



Q.34

Which of the following statement(s) is (are) correct when a mixture of $NaCl$ and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4? Mark incorrect option

(A) A deep red vapour is evolved

(B) The vapour when passed through NaOH solution, gives a yellow solution

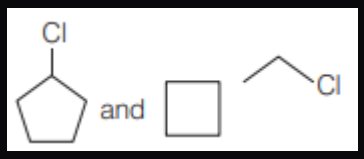
(C) Chiorine gas is also evolved

(D) Chromyl chioride is formed

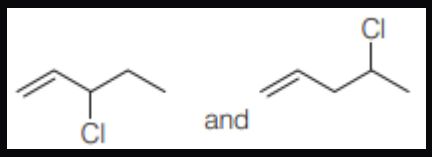
Q.35

X and Y are optically active isomers having formula C_5H_9Cl. When treated with one mole of H_2, X gets converted to an optically inactive compound Z, but Y gives an optically active compound P. The structure of X and Y respectively are

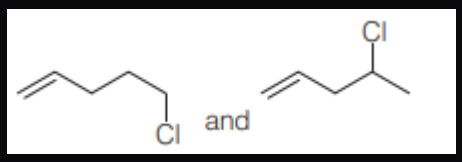
(A)



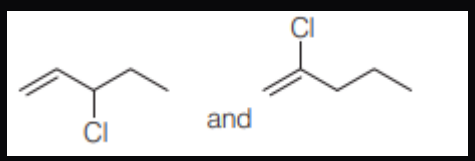
(B)



(C)



(D)



Q.36

Match the following.

List I (Element)	List II (Property)
A. He	I. Can be used as inert atmosphere
B. Ar	II. Radioactive element
C. Xe	III. Form fluoride compounds
D. Rn	IV. Can be used as airship balloons

The correct match is

$A \quad B \quad C \quad D$

(A) IV III II I

(B) IV I III II

(C) I II III IV

(D) II I IV III

Q.37

The formation of oxide ion O^{2-} from oxygen atom requires an exothermic reaction, followed by an endothermic step as shown below. The process of formation of O^{2-} in gas phase is unfavorable $(\Delta H^{\ominus} = +\text{ ve})$, even though it has stable configuration of the nearest noble gas neon. This is because

$O(g) + e^{-} \longrightarrow O^{-}(g); \Delta H^{\ominus} = -141 \text{ kJ mol}^{-1}$

$O^{-}(g) + e^{-} \longrightarrow O^{2-}(g); \Delta H^{\ominus} = +760 \text{ kJ mol}^{-1}$

(A) O^{2-} has larger size than that of neon

(B) oxygen is more electronegative than neon

(C) O^{2-} has larger size than oxygen atom

(D) electron repulsion in oxide ion is more which over comes the stability achieved by noble gas configuration.

Q.38

The green coloured complex ion of nickel in its aqueous solution is

(A) $[Ni(en)_3]^{2+}$

(B) $[Ni\,(H_2O)_2\,(en)_2]^{2+}$

(C) $[Ni(H_2O)_4(en)]^{2+}$

(D) $[Ni(H_2O)_6]^{2+}$

Q.39

For BCl_3, $AlCl_3$ and $GaCl_3$ the increasing order of ionic character is

(A) $BCl_3 < AlCl_3 < GaCl_3$

(B) $GaCl_3 < AlCl_3 < BCl_3$

(C) $BCl_3 < GaCl_3 < AlCl_3$

(D) $AlCl_3 < BCl_3 < GaCl_3$

Q.40

Assertion (A) Boron has a smaller first ionisation enthalpy than beryllium.
Reason (R) The penetration of a $2s$-electron to the nucleus is more than the $2p$-electron hence, $2p$-electron is more shielded by the inner core of electrons than 2 s-electrons.

(A) Assertion and Reason both are correct statements but Reason is not correct explanation for Assertion.

(B) Assertion is correct statement but Reason is wrong statement.

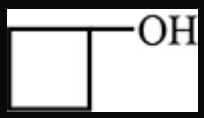
(C) Assertion and Reason both are correct statements and Reason is correct explanation for Assertion.

(D) Assertion and Reason both are wrong statements.

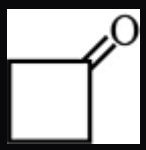
Q.41

The possible product obtained from the reaction of cyclobutyl amine with HNO_2 followed by treating with water is

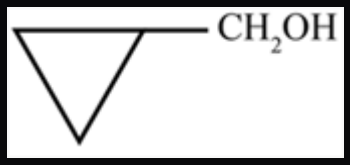
(A)



(B)



(C)



(D) Both 1 and 3

Q.42

The correct decreasing order of the basic strength is

(A) $HS^- > CH_3^- > NH_2^- > CN^-$

(B) $HS^- > CN^- > NH_2^- > CH_3^-$

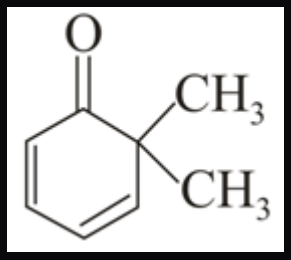
(C) $CH_3^- > NH_2^- > CN^- > HS^-$

(D) $NH_2^- > CN^- > CH_3^- > HS^-$

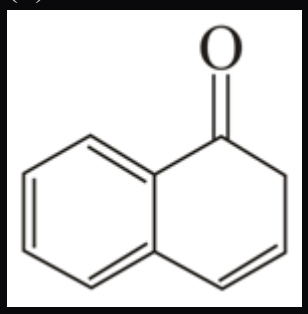
Q.43

Which of the following carbonyl compounds will exhibit enolization?

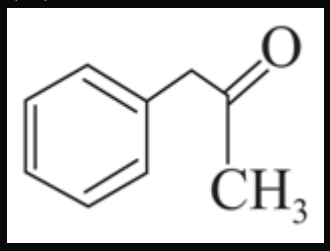
(i)



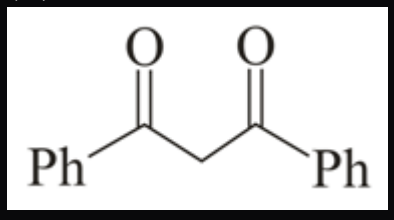
(ii)



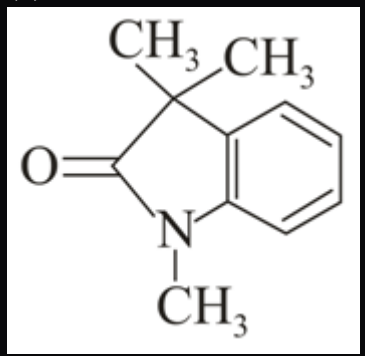
(iii)



(iv)



(v)



(A) (i), (ii) and (iii)

(B) (ii), (iii) and (iv)

(C) (iii), (iv) and (v)

(D) (i), (iii) and (v)

Q.44

Two base balls (masses: $m_1 = 100$ g, and $m_2 = 50$ g) are thrown. Both of them move with uniform velocity, but the velocity of m_2 is 1.5 times that of m_1. The ratio of de Broglie wavelengths $\lambda(m_1) : \lambda(m_2)$ is given by

(A) $4 : 3$

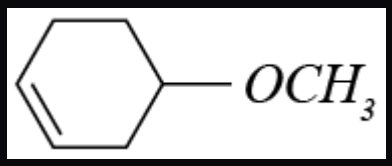
(B) $3 : 4$

(C) $2 : 1$

(D) $1 : 2$

Q.45

IUPAC name of



is

(A) 5-methoxycyclohexene

(B) methoxycyclohex-3-ene

(C) methoxycyclohex-4-ene

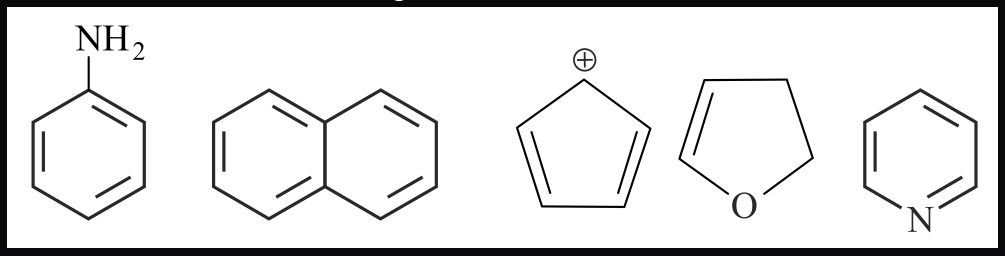
(D) 4-methoxycyclohexene

Q.46

The molar specific heat of air at room temperature and latm pressure is $25\ \text{J K}^{-1}\ \text{mol}^{-1}$. How much heat in Joule is required to heat the room through 10 Kelvin at room temperature if 144 g of air is present in room. Vapour density of air $= 14.4$.

Q.47

Total number of aromatic compounds from below is



Q.48

How many of the following orbitals have two angular and one radial nodes?
3d; 4d; 5f; 3p

Q.49

7. 8 g of a compound having molecular formula C_6H_6, on reacting with $CH_3\,COCl\,/\,AlCl_3$ gives 8. 4 g of a product which has molecular formula C_8H_8O. Calculate the percentage yield of the product C_8H_8O. (Given atomic weights of H, C and O respectively are 1, 12 and 16) (mark answer to nearest integer)

Q.50

The concentration of R in the reaction $R \rightarrow P$ was measured as a function of time and the following data is obtained:

[R](molar)	1.0	0.75	0.40	0.10
t(min).	0.0	0.05	0.12	0.18

The order of the reaction is

Q.51

If $f(x) = \lim_{n\to\infty} n^2 \left(x^{\frac{1}{n}} - x^{\frac{1}{n+1}}\right), x > 0$, then $\int xf(x)dx =$

(A) $\frac{x^2}{2}\log x + C$

(B) $\frac{x^2}{2}\log x + \frac{x^2}{4} + C$

(C) $\frac{x^2}{2}\log x - \frac{x^2}{4} + C$

(D) $-\frac{x^2}{2}\log x + \frac{x^2}{4} + C$

Q.52

Define $f(x) = \begin{cases} 1+x, & 0 \le x \le 2 \\ 3-x, & 2 < x \le 3 \end{cases}$.

If $fof(x)$ is discontinuous at a and b in $[0, 3]$ and $a < b$, then $2a + 3b =$

(A) 3

(B) 2

(C) 6

(D) 8

Q.53

Let $S = \{(a, b, c) \in N \times N \times N : a + b + c = 21, a \le b \le c\}$ and $T = \{(a, b, c) \in N \times N \times N : a, b, c$ are in $AP\}$ where N is the set of all natural numbers. Then, the number of elements in the set $\mathrm{S} \cap T$

(A) 6

(B) 7

(C) 13

(D) 14

Q.54

For an increasing A.P. $a_1, a_2, \ldots..a_n$ if $a_1 + a_3 + a_5 = -12$ and $a_1 \cdot a_3 \cdot a_5 = 80$, then which of the following is incorrect

(A) $a_1 = -10$

(B) $a_2 = -1$

(C) $a_3 = -4$

(D) $a_5 = 2$

Q.55

Let M be the foot of the perpendicular from a point P on the parabola $y^2 = 8(x - 3)$ onto its directrix and let S be the focus of the parabola. If $\triangle SPM$ is an equilateral triangle, then P is equal to

(A) $(4\sqrt{3}, 8)$

(B) $(8, 4\sqrt{3})$

(C) $(9, 4\sqrt{3})$

(D) $(4\sqrt{3}, 9)$

Q.56

Suppose the hypotenuse and its opposite vertex of an isosceles right angled triangle are $3x + 4y - 4 = 0$ and $(2, 2)$ respectively. Then, which of the following is another side of the triangle?

(A) $x - 7y - 12 = 0$

(B) $x + 7y + 12 = 0$

(C) $7x + y - 16 = 0$

(D) $7x + y + 16 = 0$

Q.57

$\sum_{k=1}^{\infty} \sum_{r=0}^{k} \frac{1}{3^k} \left({}^kC_r\right)$ is equal to

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) 1

(D) 2

Q.58

The value of the integral $\int_{-1}^{1} \left\{ \frac{x^{2013}}{e^{|x|}(x^2 + \cos x)} + \frac{1}{e^{|x|}} \right\} dx$ is equal to

(A) 0

(B) $1-e^{-1}$

(C) $2e^{-1}$

(D) $2\left(1-e^{-1}\right)$

Q.59

Let $\vec{a}=\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}$ and $\vec{b}$ be two vectors such that $\vec{a}\cdot\vec{b}=1, \cos\left(\vec{a}\cdot\vec{b}\right)=\frac{1}{3}$ and the components of $\vec{b}$ with respect to $\left(\hat{\mathrm{i}}, \hat{\mathrm{j}}, \hat{\mathrm{k}}\right)$ be integers. Then the number of possible vectors that represent $\vec{b}$ is

(A) 1

(B) 2

(C) 3

(D) 4

Q.60

For each $n \in \mathbf{N}$, let $A_n=\{(n+1)k|k \in \mathbf{N}\}$ and $X=\bigcup_{n\in\mathbf{N}} A_n \cdot A$ mapping $f: X \to N$ defined by $f(x)=x, \forall x \in \mathbf{X}$, is

(A) one-one and onto

(B) one-one but not onto

(C) onto but not one-one

(D) neither one-one nor onto

Q.61

L_1 is a line passing through the points with position vectors $\hat{\mathbf{i}}-2\hat{\mathbf{j}}-\hat{\mathbf{k}}$ and $4\hat{\mathbf{i}}-3\hat{\mathbf{k}}$. L_2 is a line passing through the points with position vectors $\hat{\mathbf{i}}+2\hat{\mathbf{j}}-\hat{\mathbf{k}}$ and $2\hat{\mathbf{i}}-4\hat{\mathbf{j}}-5\hat{\mathbf{k}}$. Then the distance between L_1 and L_2 is

(A) 0

(B) $\frac{3}{4}$

(C) $\frac{4}{3}$

(D) $\frac{2}{3}$

Q.62

Let $z=x+iy$ be a complex number,
$A=\{z||z| \leq 2\}$ and $B=\{z|(1-i)z+(1+i)\bar{z} \geq 4\}$
Then which one of the following options belongs to $A \cap B$?

(A) $\sqrt{3}+\frac{1}{2}i$

(B) $\frac{1}{2}+\frac{i}{2}$

(C) $\sqrt{2}+\frac{i}{2}$

(D) $2+2i$

Q.63

For the set $A=\{x_1,x_2,x_3,x_4,x_5\}$ the variance is 4 and the mean is 2. For the set $B=\{y_1,y_2,y_3,y_4,y_5\}$ the variance is 5 and the mean is 4. Then, the variance of $A\cup B$ is

(A) 6

(B) 6.5

(C) 5.5

(D) 5

Q.64

The foot of the perpendicular drawn from the point (1,8,4) on the line joining the point (0,-11,4) and (2,-3,1) is

(A) (4,5,2)

(B) (-4,5,2)

(C) (4,-5,2)

(D) (4,5,-2)

Q.65

The longest distance of the point $(a,0)$ from the curve $2x^2+y^2=2x$ is

(A) $1+a$

(B) $|1-a|$

(C) $\sqrt{1-2a+2a^2}$

(D) $\sqrt{1-2a+3a^2}$

Q.66

The particular solution of the different equation
$\frac{dx}{dy}=\frac{\sin y(1+y\cot y)}{x\log(x^2e)}, y(1)=0$

(A) $y\sin y=x^2\log x$

(B) $y^2\sin y=\log x$

(C) $y=\left(\frac{e^2}{\sin e}\right)(x-1)$

(D) $y=e^2\sec x$

Q.67

The distinct points $A(0,\ 0),\ B(0,\ 1),\ C(1,\ 0)$ and $D(2a,\ 3a)$ are concyclic, then

(A) 'a' can attain only rational values

(B) 'a' is irrational

(C) Cannot be concyclic for any 'a'

(D) None of the above

Q.68

There are two possible values of A in the solution of the matrix equation $\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$ where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions of A is

(A) $\frac{8}{3}$

(B) $\frac{11}{3}$

(C) $\frac{1}{3}$

(D) $\frac{19}{3}$

Q.69

If L and M are respectively the coefficient of x^{-7} in $\left(ax+\frac{b}{x^2}\right)^{11}$ and the coefficient of x^7 in $\left(bx^2+\frac{a}{x}\right)^{11}$ then $L+M=$

(A) $\frac{1}{b}\left[\text{coefficient of } x^{-6} \text{ in } \left(ax+\frac{b}{x^2}\right)^{12}\right]$

(B) $\frac{1}{a}\left[\text{coefficient of } x^{6} \text{ in } \left(ax^2+\frac{b}{x}\right)^{12}\right]$

(C) $a\left[\text{coefficient of } x^{-10} \text{ in } \left(ax+\frac{b}{x^2}\right)^{11}\right]$

(D) $b\left[\text{coefficient of } x^{4} \text{ in } \left(ax^2+\frac{b}{x}\right)^{11}\right]$

Q.70

Let $\mathrm{F(x)}$ be a non-negative continuous function defined on R such that $\mathrm{F(x)+F\left(x+\frac{1}{2}\right)=1}$. Then value of $\frac{1}{1000}\int_0^{1500} F(x)dx$

(A) 0.5

(B) 1.5

(C) 0.75

(D) 0.25

Q.71

Let $\overrightarrow{\mathrm{a}}=\hat{\mathrm{i}}+2\hat{\mathrm{j}}+3\hat{\mathrm{k}}$ and $\overrightarrow{\mathrm{b}}=\hat{\mathrm{i}}-2\hat{\mathrm{j}}-3\hat{\mathrm{k}}$ be two vectors. If A_1 is the area of the quadrilateral having $\vec{a}, \vec{b}$ as its diagonals and A_2 is the area of the parallelogram having $\overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{b}}$ as its two adjacent sides, then $\mathrm{A}_1 \cdot \mathrm{A}_2=$

Q.72

There are two dice A and B. Die A has 4 red and 2 white faces and B has 2 red and 4 white faces. A coin is tossed once, if it shows head, die A is rolled, if it shows tail, die B is rolled, if the probability that die A is used is $\left(\frac{32}{33}\right)$ when it is given that red turns up every time in first n throws, then $n=$

Q.73

Let $A_r=\left(x+\frac{1}{x}\right)^3 \cdot\left(x^2+\frac{1}{x^2}\right)^3 \cdot\left(x^3+\frac{1}{x^3}\right)^3 \ldots .\left(x^r+\frac{1}{x^r}\right)^3$. If $x^2+x+1=0$, then $\frac{1}{A_3}+\frac{1}{A_6}+\frac{1}{A_9}+\frac{1}{A_{12}}+\ldots . .\infty=K$, find 4/K.

Q.74

$OABC$ is a unit square where O is the origin and $B=(1,1)$. The curves $y^2=x$ and $x^2=y$ divide the area of the square into three parts $OABO, OBO$ and $OBCO$. If a_1, a_2, a_3 are the areas (in sq units) of these parts respectively, then $a_1+2a_2+3a_3=$

Q.75

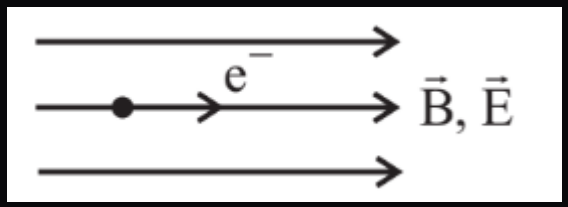
a, b, c, d, e, f, g, h be distinct elements in the set $\{-7,-5,-3,-2,2,4,6,13\}$. The minimum value of $(a+b+c+d)^2+(e+f+g+h)^2$ is

Answers & Solutions

Q.1 Answer:

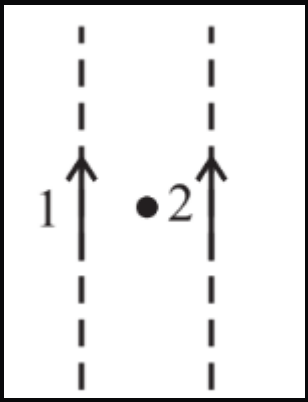
—

Solution:



$\overrightarrow{\mathrm{B}}$ **will not effect electron and** $\overrightarrow{\mathrm{E}}$ **will reduce electron velocity as electric field on it is in direction opposite to field.**

So (I) is correct



$\overrightarrow{\mathrm{B}}_{\mathrm{net}}=\frac{\mu_0 \mathrm{i}}{2 \pi \mathrm{r}} \otimes+\frac{\mu_0 \mathrm{i}}{2 \pi \mathrm{r}} \odot=0$

So, (II) is correct

No net force acts on rectangular loop placed in magnetic field.

So (III) is correct.

Q.2 Answer:

$2.1 \mathrm{~V}$

Solution:

The first spectral line corresponds to transition from $n=2$ to $n=1$.

Therefore, the energy of first excitation of sodium will be

$$E=h v=\frac{h c}{\lambda}$$

Where h is Planck's constants, v is frequency, c is speed of light and λ is wavelength.

$$E=\frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{5896 \times 10^{-10}}$$

$$E=3.37 \times 10^{-19} \mathrm{~J}$$

Also since $1.6 \times 10^{-19} \mathrm{~J}=1 \mathrm{eV}$

$$\therefore \quad E=\frac{3.37 \times 10^{-19}}{1.6 \times 10^{-19}} \mathrm{eV}$$

$$E=2.1 \mathrm{eV}$$

Hence, corresponding first excitation potential is $2.1 \mathrm{~V}$.

Q.3 Answer:

128

Solution:

$\gamma=\frac{7}{5}$

For adiabatic process $PV^{\gamma}=$ constant where,

$V=$ volume

$\therefore \quad Pd^{-7}=$ constant

$\left(\because v \propto \frac{1}{d}\right)$

where d = density

$$P_1 d_1^{-7} = P_2 d_2^{-\gamma}$$

$$\frac{P_1}{P_2} = \left(\frac{d_2}{d_1}\right)^{-\gamma} \Rightarrow \frac{P_1}{P_2} = \left(\frac{d_1}{d_2}\right)^{\gamma}$$

$$\frac{P}{P'} = \left(\frac{d}{d'}\right)^{7/5} \Rightarrow \frac{P'}{P} = \left(\frac{d'}{d}\right)^{7/5}$$

$$= (32)^{7/5}$$

$$= \left(2^5\right)^{7/5} = 2^7 = 128$$

Q.4 Answer:

B and C false, A is true

Solution:

Statement (A): When the temperature increases the viscosity of gases increases and the viscosity of liquids decreases.

Statement (B): Water does not wet an oily glass because cohesive force of oil is less than that of water.

Statement (C) : A liquid will wet a surface of a solid if the angle of contact is greater than 90°.

Q.5 Answer:

$d/(m+1)$ unit

Solution:

$(m+1)$ vernier division $= m$ main scale division One division on vernier scale $= \left(\frac{m}{m+1}\right)$ division on main scale Vernier constant $= \left(1 - \frac{m}{m+1}\right) d = \frac{d}{m+1}$

Q.6 Answer:

$\frac{H}{3}, 2\sqrt{\frac{gH}{3}}$

Solution:

If particle falls by a distance x, then

$$\mathrm{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m(2gx) = mgx$$

$$\mathrm{PE} = mg(H - x)$$

As, $\mathrm{KE} = 2(\mathrm{PE}) \Rightarrow mgx = 2mg(H - x) \Rightarrow x = \frac{2H}{3}$

So, height of particle is $H - x = \frac{H}{3}$.

Speed of particle at $\frac{H}{3}$ distance $= \sqrt{2g\left(\frac{2H}{3}\right)} = 2\sqrt{\frac{gH}{3}}$

Q.7 Answer:

$20\pi \mathrm{mV}$

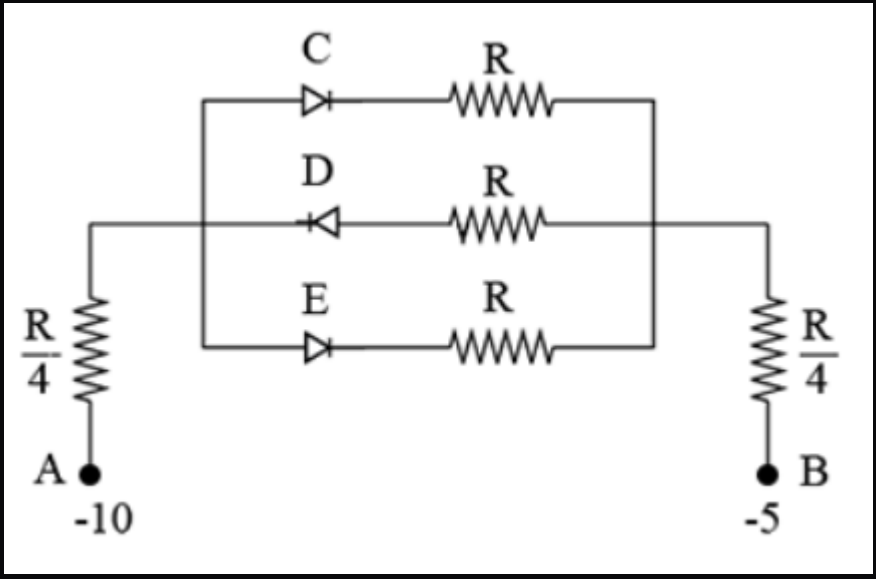
Solution:

$$\varepsilon = \frac{B\omega \mathrm{R}^2}{2}$$
$$= \frac{0.1 \times (2\pi \times 20)(0.1)^2}{2}$$
$$= 20\pi \times 10^{-3}\ \mathrm{V}$$

Q.8 Answer:

(ii) $=$ (iii) $<$ (i)

Solution:



i) In this case of current will flow from B top A. No current will flow through branch C, E (as marked in figure)

$\Rightarrow \quad R_{\text{eq}} = \frac{R}{4} + R + \frac{R}{4} = \frac{3R}{2}$

ii) Here current will flow from A to B so no current will flow in branch D

$\Rightarrow \quad R_{\text{eq}} = \frac{R}{4} + \frac{R}{2} + \frac{R}{4} = R$

iii) Here also current will flow from A to B, so $R_{\text{eq}} = R$ same as (ii)

$\Rightarrow \quad R_{\text{ii}} = R_{\text{iii}} < \mathrm{R_i}$

Q.9 Answer:

$\sqrt{5} : \sqrt{12}$

Solution:

Speed of sound in gas

$$v=\sqrt{\frac{\gamma RT_1}{M}} \quad \ldots(\mathrm{i})$$

rms speed of molecules,

$$c=\sqrt{\frac{3RT_2}{M}} \quad \ldots(\mathrm{ii})$$

For monoatonic gas, $r=\frac{5}{3}$

Also, $T_1=300\ \mathrm{K}, T_2=400\ \mathrm{K},$

$$\therefore \quad \text{Ratio, } \frac{v}{c}=\sqrt{\frac{\gamma \cdot \mathrm{T}_1}{3 \cdot T_2}}=\sqrt{\frac{5/3 \times 300}{3 \times 400}}$$

$$\frac{v}{c}=\frac{\sqrt{5}}{\sqrt{12}}$$

Q.10 Answer:

25

Solution:

$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3} \Rightarrow f_{eq}=1\ \mathrm{cm}$

$$m=\begin{cases} \frac{D}{fe}, & \text{when final image is at } \infty \text{ (Normal Adj.)} \\ 1+\frac{D}{fe}, & \text{Max. Magnification} \end{cases}$$

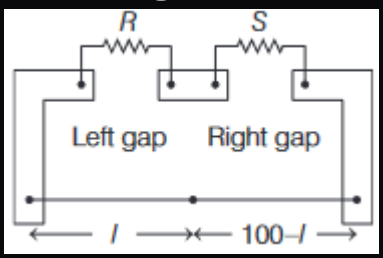
$$m=\frac{25}{1}=25$$

Q.11 Answer:

$100 \cdot \frac{n^2}{n^2+4}$

Solution:

Meter bridge is shown in the figure below,



When $\frac{n}{2}$ **resistances are joined in series in left gap each of resistance** R_1**, then equivalent resistance in left gap.**

$R=\frac{R_1}{2}+\frac{R_1}{2}+\frac{R_1}{2}+\ldots \frac{n}{2} \cdot \text{ times } =\frac{R_1 n}{2}$

When $\frac{n}{2}$ **resistors are joined in parallel in right gap, then the equivalent resistance in right gap.**

$$\frac{1}{S}=\frac{1}{R_1}+\frac{1}{R_1}+\frac{1}{R_1}+\ldots . \frac{n}{2} \text{ times } =\frac{n}{2R_1}$$

$$\Rightarrow \quad S=\frac{2R_1}{n}$$

If l be the balancing length in the meter bridge wire, then

$$\frac{R}{S}=\frac{l}{100-l} \Rightarrow \frac{\frac{R_1 n}{2}}{\frac{2R_1}{n}}=\frac{l}{100-l}$$

$$\frac{n^2}{4}=\frac{l}{100-l} \Rightarrow l=\frac{100n^2}{n^2+4}$$

Q.12 Answer:

0.5 mm

Solution:

We have

$$|\mathrm{E}|=\frac{\Delta \mathrm{V}}{\Delta \mathrm{r}}=\frac{19.8}{\Delta \mathrm{r}}$$

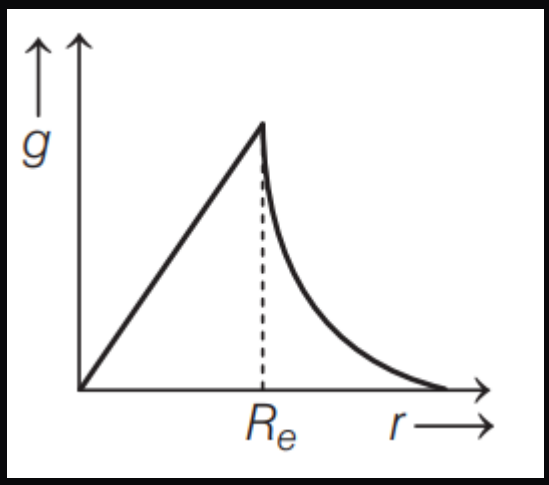
$$\Rightarrow \frac{\sigma}{2\varepsilon_0}=\frac{19.8}{\Delta \mathrm{r}}$$

$$\Rightarrow \Delta \mathrm{r}=\frac{19.8 \times 2\varepsilon_0}{\sigma}=\frac{19.8 \times 2 \times 8.85 \times 10^{-12}}{7 \times 10^{-7}}$$

$$=5 \times 10^{-4} \text{ m}$$

$$=0.5 \text{ mm}$$

Q.13 Answer:

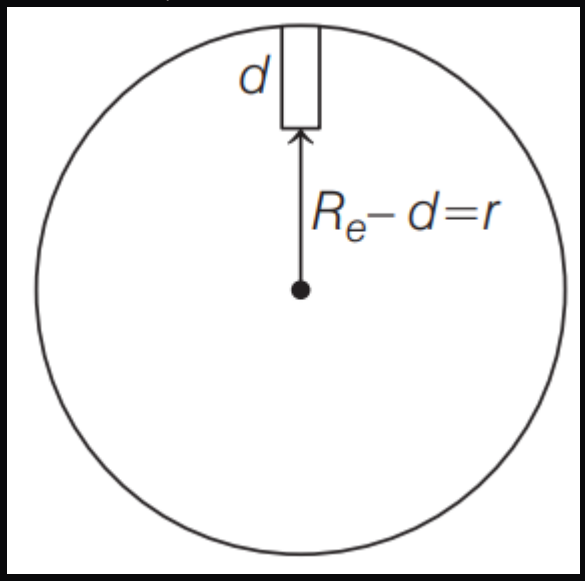


Solution:

At depth d,

$$g'=g\left(1-\frac{d}{R_e}\right)=g\left(\frac{R_e-d}{R_e}\right)=\frac{gr}{R_e}$$

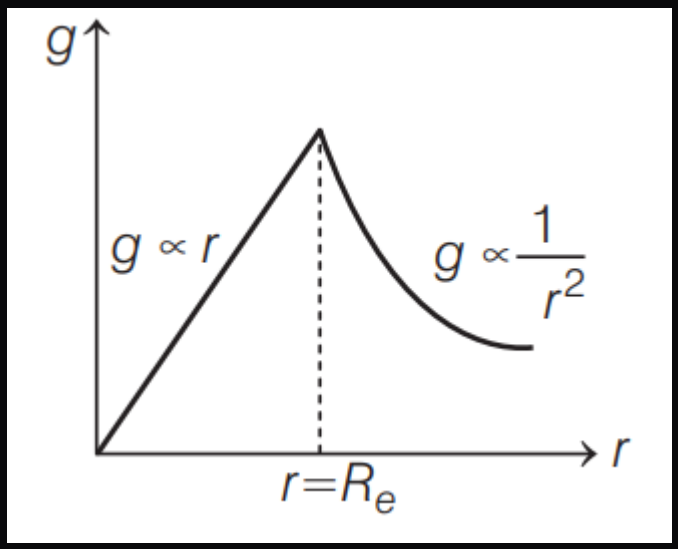
$\Rightarrow g' \propto r$(where, r = distance from centre of earth)



At height h,

$g' = g\left(\frac{R_e^2}{(R_e+h)^2}\right) = \frac{gR_e^2}{r^2} \Rightarrow g' \propto \frac{1}{r^2}$

So, correct graph is

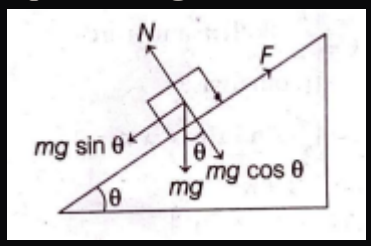


Q.14 Answer:

m_1, m_2 and m_3 begins to slide at the same inclination angle.

Solution:

An object started to slide down an inclined plane when angle of incline (also called angle of repose) is just equals to angle of friction. Let θ = angle at which a mass started to slide down an incline.



Then, force of friction = downward force .

$\Rightarrow \mu N = mg\sin\theta$

$\Rightarrow \mu mg\cos\theta = mg\sin\theta$

$\Rightarrow \quad \tan\theta = \mu$

$\Rightarrow \qquad \theta = \tan^{-1}\mu$

Hence, angle does not depend on mass of body. So all three masses slides down at same instant at same inclination angle (θ).

Q.15 Answer:

$G^{\frac{1}{2}}\, h^{\frac{1}{2}}\, c^{\frac{-5}{2}}$

Solution:

Let time, $T \propto c^X G^Y h^Z$

$\Rightarrow T = kc^X G^Y h^Z$

Taking dimensions on both sides

$[M^0L^0T^1] = [LT^{-1}]^X [M^{-1}L^3T^{-2}]^Y [ML^2T^{-1}]^Z$

i.e.,

$[M^0L^0T^1] = [M^{-Y+Z}L^{X+3Y+2Z}T^{-X-2Y-Z}]$

Equating powers of $M,\ L$, and T, on both sides, we get –

$-Y + Z = 0 \quad ...(1)$

$X + 3Y + 2Z = 0 \quad ...(2)$

$-X - 2Y - Z = 1 \quad ...(3)$

From equation (i)

Adding (2) and (3)

$Y + Z = 1$

or $\qquad 2Y = 1 \quad (\because\ Z = Y)$

$\therefore Z = Y = \frac{1}{2}$

Putting these values in (2), we get –

$X + \frac{3}{2} + 1 = 0$ or $X = \frac{-5}{2}$

Hence, $[T] = [G^{\frac{1}{2}}\, h^{\frac{1}{2}}\, c^{\frac{-5}{2}}]$

Q.16 Answer:

$5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{5} - 6 \times 10^{+7} t\right)\right] \hat{z}\ \mathrm{T}$

Solution:

$$B_0 = \frac{E_0}{C} = \frac{150}{3 \times 10^8} = 5 \times 10^{-7}$$

$$f = 30\mathrm{MHz} = 30 \times 10^6\ \mathrm{Hz}$$

$$\omega = 60\pi \times 10^6\ \mathrm{Hz},\ \mathrm{K} = \frac{\omega}{\mathrm{C}} = \frac{\mathrm{x}}{5}$$

as $\hat{\mathrm{C}} = \overrightarrow{\mathrm{E}} \times \overline{\mathrm{B}}$

So, direction of $\overrightarrow{\mathrm{B}}$ **is along z-axis.**

$$\text{Thus, } \overrightarrow{\mathrm{B}} = \left(5 \times 10^{-7}\right) \sin\left[\frac{\pi}{5}\mathrm{x} - 6 \times 10^7 \pi \mathrm{t}\right] \hat{z}$$

Q.17 Answer:

If both Assertion and Reason are true but Reason is not the explanation of Assertion.

Solution:

Superconductivity is a phenomenon of exactly zero electrical resistance and expulsion of magnetic flux fields occurring in certain materials. These materials are called superconductors.
In a superconductor, below 'critical temperature', the electrical resistance suddenly falls to zero. As electrical resistance becomes zero, there is no loss of energy during power transmission.
So, Assertion and Reason are both true but Reason does not explain Assertion.
The actual explanation is beyond the scope of JEE/NEET syllabus, but still, we can have some idea. On a microscopic scale, according to Quantum Physics, electrons behave like small waves. When there is a flaw or when one of the atoms of the crystal pattern is vibrating, the wave gets disrupted. At a very low temperature, the electrons pair up and merge into one quantum wave that fills the whole material. This unique wave becomes insensitive to the flaws in the material. They are too small to slow the whole wave. The electric resistance has hence disappeared.

Q.18 Answer:

60°

Solution:

$$\mu = \sqrt{3}, \delta_m = A$$

$$\text{For prism } \mu = \frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$$

$$\sqrt{3} = \frac{\sin\frac{A+A}{2}}{\sin\frac{A}{2}}$$

$$\sqrt{3} = \frac{\sin A}{\sin\frac{A}{2}} \Rightarrow \sqrt{3} = \frac{2\sin\frac{A}{2}\cos\frac{A}{2}}{\sin\frac{A}{2}}$$

$$\cos\frac{A}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \quad \cos\frac{A}{2} = \cos 30^\circ$$

$$\therefore \quad \frac{A}{2} = 30^\circ \Rightarrow A = 60^\circ$$

Q.19 Answer:

(A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

Solution:

Fleming's left hand rule	—	Direction of force due to magnetic induction.
Right hand thumb rule	—	Direction of magnetic lines due to current.

Biot-savart law	—	Magnitude and direction of magnetic induction.
Fleming's right hand rule	—	Direction of induced current.

So, the option (b) is correct.

Q.20 Answer:

$\frac{2}{3}$

Solution:

Tension between P and Q is

$T_1 =$ **centripetal force on** $Q = mr\omega^2$

$= 200 \times 1 \times \omega^2 \left(\mathrm{g \cdot m \cdot \frac{rad^2}{s^2}} \right)$

Tension between O and P is

$T_2 =$ **centripetal force on** $Q +$ **centripetal force on** P

$= 200 \times 1 \times \omega^2 + 200 \times \frac{1}{2} \times \omega^2$

$= 300 \times 1 \times \omega^2 \left(\mathrm{g \cdot m \cdot \frac{rad^2}{s^2}} \right)$

Ratio of tension is

$\frac{T_1}{T_2} = \frac{200 \times 1 \times \omega^2}{300 \times 1 \times \omega^2} = \frac{2}{3}$

Q.21 Answer:

—

Solution:

The energy stored in an inductor is given by the formula

$U = \frac{1}{2} L i^2$

Given, $U = 0.1 \text{ J}, i = 0.2 \text{ A}$

$$0.1 = \frac{1}{2} L \times (0.2)^2$$

$$\Rightarrow L = \frac{2 \times 0.1}{(0.2)^2} = \frac{0.2}{0.2 \times 0.2}$$

$$= \frac{10}{2} = 5\text{H}$$

Q.22 Answer:

—

Solution:

Radius of the loop, $r = 1 \text{ cm} = 0.01 \text{ m}$

Total charge on the loop, $Q = 1 \times 10^{-6}\text{C}$

$$\therefore \text{ Charge per unit length, } \lambda = \frac{Q}{2\pi r} = \frac{10^{-6}}{2\pi \times 0.01}$$

$$= \frac{10^{-4}}{2\pi}\text{C/m}$$

$$l' = 0.01\% \text{ of the length } = \frac{0.01}{100} \times 2\pi r$$

$$\Rightarrow \quad l' = \frac{2\pi \times 0.01 \times 0.01}{100}$$

$$= 2\pi \times 10^{-6} \text{ m}$$

For the whole circular loop, the electric field at the centre is zero due to symmetry. But when a certain portion of the loop is cut off, then there will exist an electric field at the centre due to that cutting part. Which is equal to the electric field due to remaining part of wire as per charge conservation.

∴ Charge of the part $l', Q' = l'\lambda = \frac{2\pi \times 10^{-6} \times 10^{-4}}{2\pi}$

$= 10^{-10}\text{C}$

∴ Electric field at centre,

$$E = \frac{Q'}{4\pi\varepsilon_0 r^2}$$

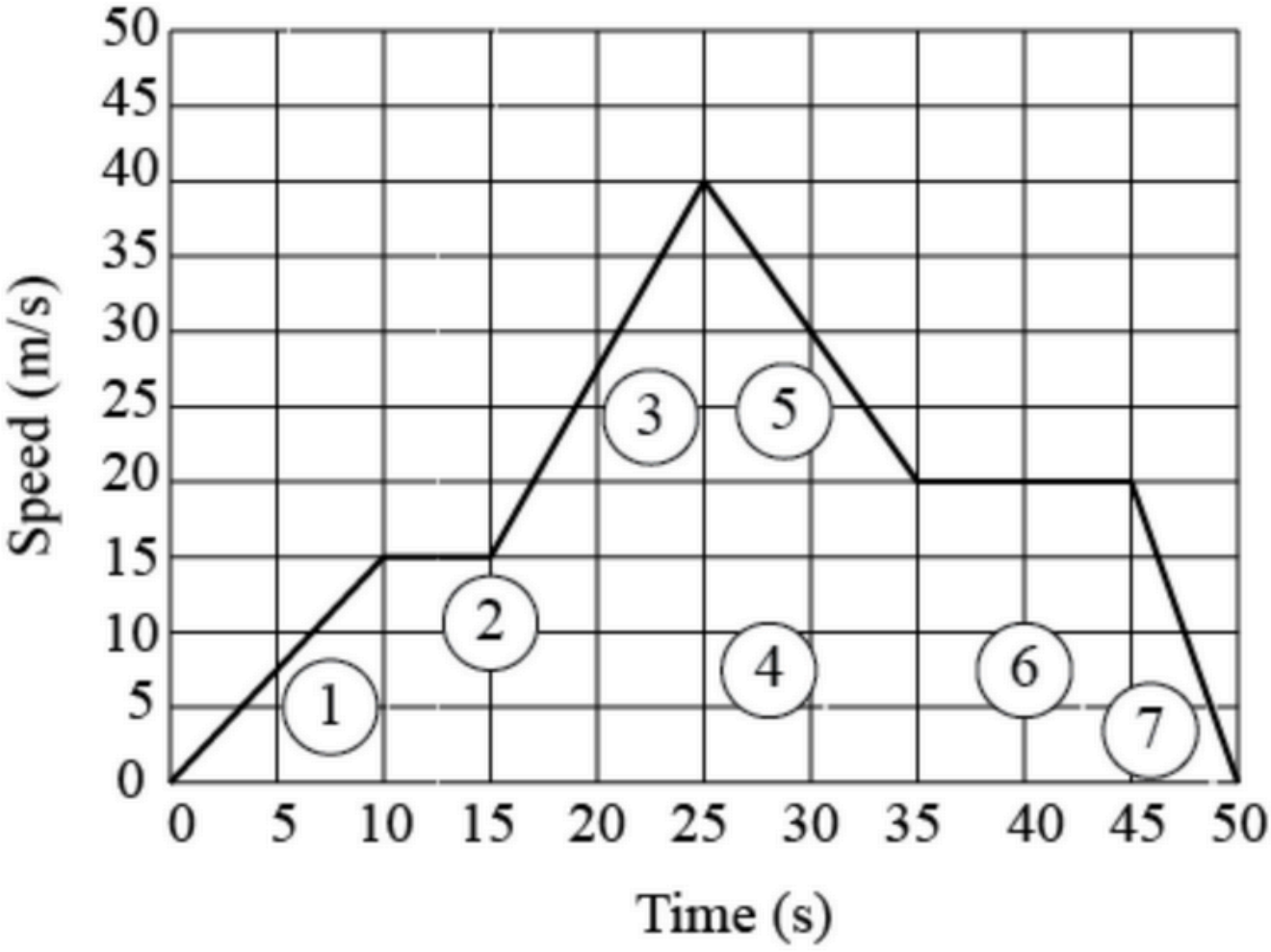
$$E = \frac{10^{-10} \times 9 \times 10^9}{(0.01)^2}$$

$$= 9 \times 10^3 \text{ N/C}$$

Q.23 Answer:

—

Solution:



The total distance travelled by the object will be area of given graph in the figure.

Total distance $s = 75 + 225 + 125 + 100 + 400 + 50 = 975 \text{ m}$

Q.24 Answer:

—

Solution:

Given nuclear reaction is

$^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He} + \text{Q}$

Mass defect $= M_{\text{U}} - M_{\text{Th}} - M_{\text{He}}$

= 238.05079 – 234.043630 – 4.002600

= 0.00456 u

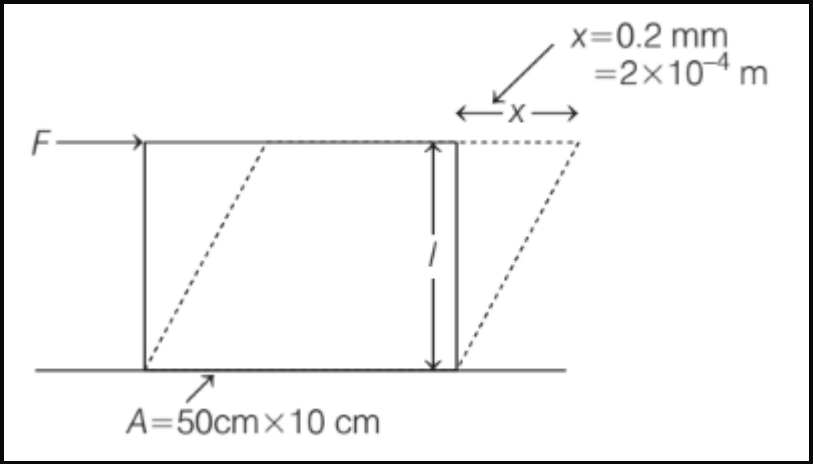
Energy released = $(0.00456 \text{ u}) \times (931.5 \text{ MeV}) = 4.25 \text{ MeV}$ **.**

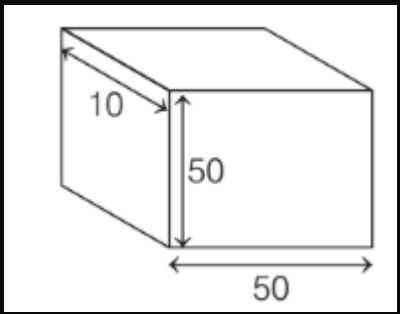
Q.25 Answer:

—

Solution:

The given situation is shown in the following figure.





$$\text{Shear strain} = \frac{x}{l} = \frac{0.2\text{ mm}}{50\text{ cm}} = \frac{0.2\times10^{-3}}{50\times10^{-2}}$$

$$\text{Shear stress} = \frac{F}{A} = \frac{10^5}{\frac{50}{100}\times\frac{10}{100}} = \frac{10^5\times10^4}{5\times10^2} = 2\times10^6$$

$$\text{Modulus of rigidity} = \frac{\text{Stress}}{\text{Strain}} = \frac{2\times10^6\times50\times10^{-2}}{0.2\times10^{-3}}$$

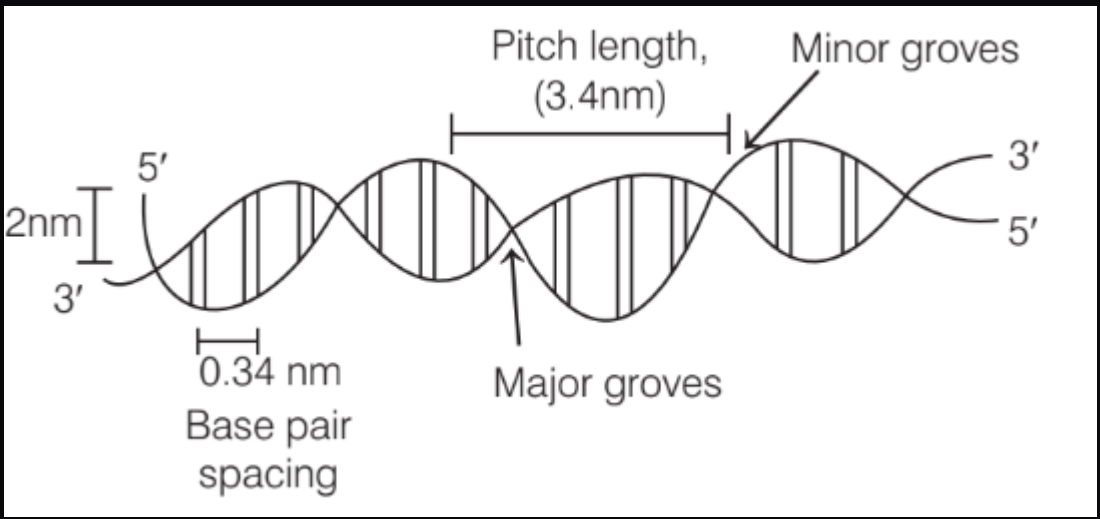
$$= 5\times10^9\ \text{N/m}^2 = 5\text{GPa}$$

Q.26 Answer:

The two strands in a DNA molecule are not complementary to each other

Solution:

I. DNA has a double helix structure.



II. Adenine form hydrogen bond with thymine $A = T$) and cytocine forms hydrogen bond with guanine $(C \equiv G)$.

III. The two strands in DNA molecules are complementary to each other.

IV. DNA strand contains pentose -2-deoxyribose sugar.

Thus, option (c) is inorrect.

Q.27 Answer:

iv $<$ ii $<$ iii $<$ i

Solution:

The order of increase of energy can be calculated from $(n+I)$ rule. If two orbitals have same value of $(n+I)$, the orbital with lower value of n will be filled first.

(i) For $n=4, I=1, (n+I)=4+1=5$

(ii) For $n=4, l=0, (n+l)=4+0=4$

(iii) For $n=3, l=2, (n+l)=3+2=5$

(iv) For $n=3, I=1, (n+I)=3+1=4$

Therefore correct order is

(iv) $<$ (ii) $<$ (iii) $<$ (i).

Q.28 Answer:

Both (A) and (R) are true and (R) is not the correct explanation of (A)

Solution:

Both (A) and (R) are true statement but (R) is not the correct explanation of (A). pH of a buffer solution is given by Henderson-Haselbalch equation.

$$\mathrm{pH}=\mathrm{p}K_a+\log\frac{[\text{ salt }]}{[\text{acid }]}$$

$$\therefore \quad \mathrm{pH}=\mathrm{p}K_a(\text{ when }[\text{ salt }]=[\text{ acid }])$$

Q.29 Answer:

(A) is true but (R) is false

Solution:

Tertiary alcohols produce alkyl halides with Lucas reagent (conc. $\mathrm{HNO_3}+\mathrm{ZnCl_2}$), which is immiscible. $\mathrm{ZnCl_2}$ acts as catalyst.

Q.30 Answer:

Eu, Yb

Solution:

Both Eu^{+2} and Yb^{+2} tends to acquire common oxidation state of +3 by loosing 1 electron and becomes highly stable.

Q.31 Answer:

6

Solution:

Relative lowering of vapour pressure,

$$\frac{p^\circ - p_s}{p^\circ} = x_A$$

$$= \frac{\frac{w_A}{m_A}}{\frac{w_A}{m_A} + \frac{w_B}{m_B}}$$

(where, w_A and m_A are the mass and molar mass of solute and w_B and m_B are the mass and molar mass of water.)

$$0.02 = \frac{\frac{x}{60}}{\frac{x}{60} + \frac{90}{18}}$$

$$0.02 = \frac{\frac{x}{60}}{\frac{x}{60} + 5}$$

$$\therefore \quad \frac{1}{0.02} = \frac{\frac{x}{60} + 5}{\frac{x}{60}}$$

$$50 = 1 + \frac{5 \times 60}{x}$$

$$49 = \frac{300}{x}$$

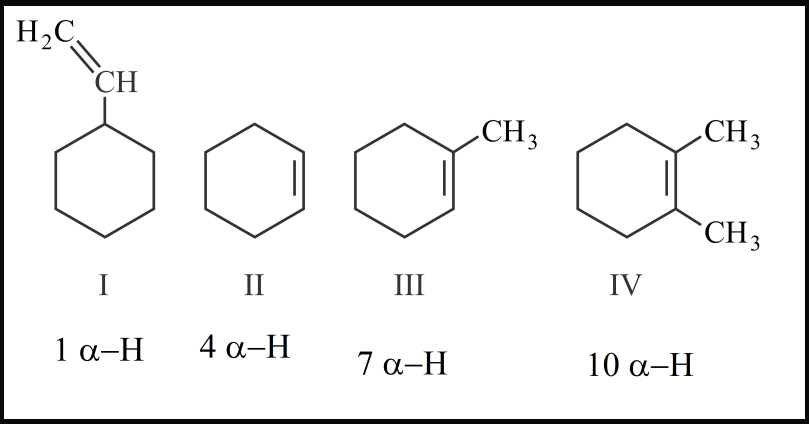
$$x = \frac{300}{49} = 6 \text{ g}$$

Q.32 Answer:

IV > III > II > I

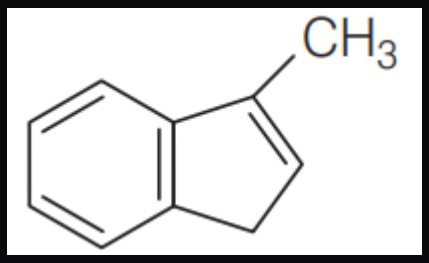
Solution:

Hyperconjugation increases the stability of alkene which does not have any other resonance effect. And hyperconjugation effect depends on the number of alpha hydrogen around the alkene. More the number of alpha hydrogen atoms more stable the alkene will be.



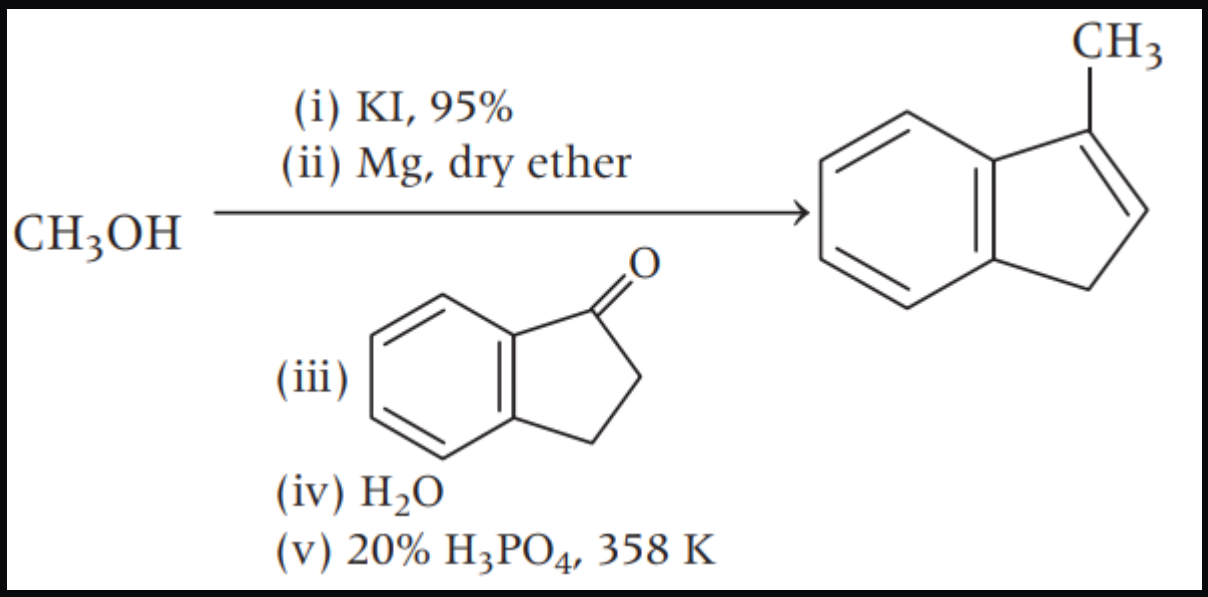
So, stability order will be IV > III > II > I.

Q.33 Answer:



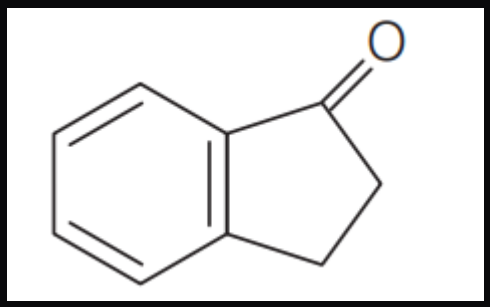
Solution:

The given chain reaction give the final product as follows :



(i) with KI, we get CH_3I, which on treatment with Mg, dry ether give ethane $(CH_3 - CH_3)$.

(ii) on treatment with



followed by H_2O and $20\% H_3PO_4$ at $358\ K$, we get the final product.

Q.34 Answer:

Chiorine gas is also evolved

Solution:

(A) When $NaCl$ and $K_2Cr_2O_7$ warmed with H_2SO_4 (i.e. in acidic medium), they produce deep red vapours of chromyl chloride (CrO_2Cl_2)

(B) When NaOH is passod, the product CrO_3, formed In step (a) will react with NaOH and gives yellow colour solution

(C) Chlorine gas is not evolved, thus false

(D) In the given reaction, chromyl chloride (CrO_2Cl_2) is formed. All reactions are as follows

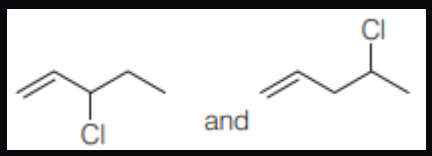
(i) $K_2Cr_2O_7 + 2H_2SO_4 \longrightarrow 2KHSO_4 + 2CrO_3 + H_2O$

(ii) $NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$

(iii) $CrO_3 + 2HCl \longrightarrow CrO_2Cl_2$

(iv) $CrO_2Cl_2 + 2NaOH \longrightarrow Na_2CrO_4 + 2HCl$

Q.35 Answer:

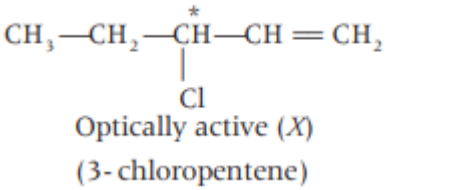


Solution:

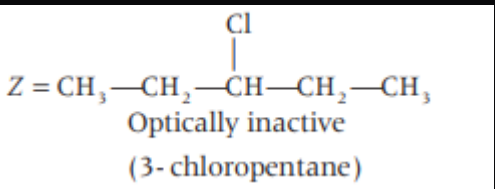
Let us consider the formula of C_5H_9Cl which is optically active.

Optically active means molecule does not contain plane of symmetry.

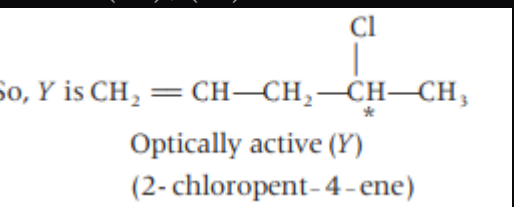
So, possible optically inactive structure of C_5H_9Cl is



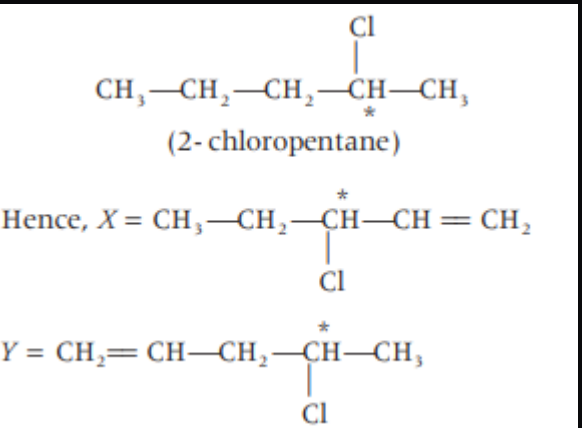
When, ' X' react with 1 mole of H_2, it converted into optically inactive compound ' Z '



Same as $(X), (Y)$ is optically active and give compound (P) which is inactive.



When ' Y ' react with 1 mole of H_2, it will give ' P ' which is optically active. So, ' P ' is



Q.36 Answer:

IV I III II

Solution:

(A) Helium is lighter than air, so used in airship balloons.
(B) Argon is completely inert to other elements, so used as inert atmosphere.
(C) Due to larger size of Xe, the outer electron and nucleus attraction is weak. So, it reacts with highly electronegative element, i.e. fluorine and oxygen.
D) Radon is formed from the natural decay of uranium and radium, so, it is radioactive.
The correct match is A-IV, B-I, C-III and D-II.

Q.37 Answer:

electron repulsion in oxide ion is more which over comes the stability achieved by noble gas configuration.

Solution:

The process of formation of O^{2-} in the gas phase is unfavorable even through O^{2-} is isoelectronic with neon. It is due to that electronic repulsion outweighs the stability gained by achieving noble gas configuration. The formation of oxide ion, $O^{2-}(g)$ from oxygen atom require first an exothermic and then endothermic step.

$$O(g) + e^- \longrightarrow O^-(g); \Delta H^\ominus = -141 \text{ kJ mol}^{-1}$$

$$O^-(g) + e^- \longrightarrow O^{2-}(g); \Delta H^\ominus = +760 \text{ kJ mol}^{-1}$$

When an electron is added to O^- anion, there is strong electrostatic repulsion between two negative charge due to this, the second electron gain enthalpy of oxygen is positive.

Q.38 Answer:

$[Ni(H_2O)_6]^{2+}$

Solution:

In complex $[Ni(H_2O)_6]^{2+}$ the strong ligand is absent. Thus, electrons do not go for pairing and two unpaired e^- are present in Ni^{2+} complex. Hence, is a coloured complex.

Q.39 Answer:

$BCl_3 < GaCl_3 < AlCl_3$

Solution:

Since, ionic character is inversely proportional to polarising power of cation therefore correct order is $AlCl_3 > GaCl_3 > BCl_3$

Q.40 Answer:

Assertion and Reason both are correct statements and Reason is correct explanation for Assertion.

Solution:

Assertion (A) Electronic configuration of boron $(Z = 5)$ is [He] $2s^2 2p^1$. So, in first ionisation ($1IE_1$ or $\Delta_i H_1$) removal will take place from unpaired p^1-electron. Whereas that of Be will be from paired $2s^2$-

electrons which requires more energy.

Electronic configuration of $\mathrm{Be}(Z=4):[\mathrm{He}]2s^2$ So, the Assertion is a correct statement.

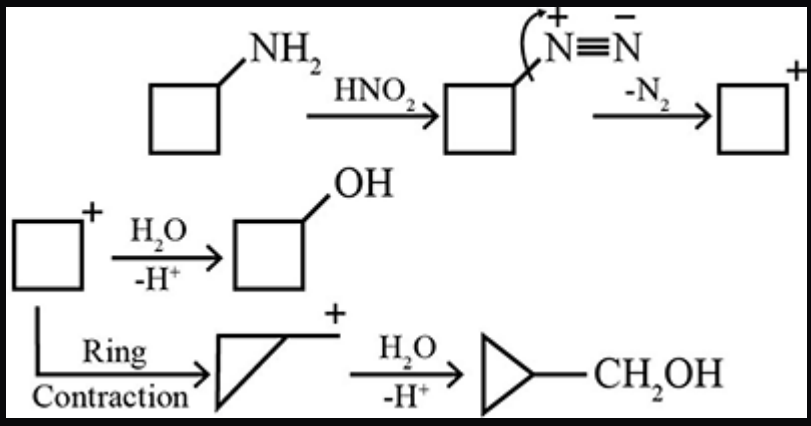
Reason (R) s-orbital is being symmetrical in shape (spherical), it shields nuclear force (nuclear charge) strongly. So, $2p^1$-electron of B is experiences lesser nuclear attractive force for ionisation. As a result, IE_1 or $\Delta_i H_1 : \mathrm{B} < \mathrm{Be}$

So, the Reason is correct explanation for Assertion.

Q.41 Answer:

Both 1 and 3

Solution:



Q.42 Answer:

$\mathrm{CH_3^-} > \mathrm{NH_2^-} > \mathrm{CN^-} > \mathrm{HS^-}$

Solution:

According to the question, Conjugate acids and bases are part of the Bronsted-Lowry theory of acids and bases.

According to this theory, the species that donates a hydrogen cation or proton in a reaction is a conjugate acid, while the remaining portion or the one that accepts a proton or hydrogen is the conjugate base.

The conjugate base may be recognised as an anion.

Conjugated base of weak acid is worked as strong base.

$\mathrm{CH_3^-} + \mathrm{H^+} \rightarrow \mathrm{CH_4}\ (\mathrm{W.A})\ \mathrm{CN^-} + \mathrm{H^+} \rightarrow \mathrm{HCN}\ \mathrm{NH_2^-} + \mathrm{H^+} \rightarrow \mathrm{NH_3}\ \mathrm{HS^-} + \mathrm{H^+} \rightarrow \mathrm{H_2S}\ (\mathrm{S.A})$

Q.43 Answer:

(ii), (iii) and (iv)

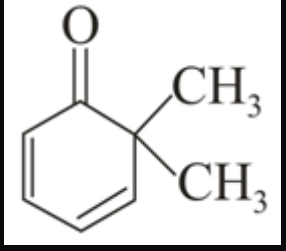
Solution:

For the enolization carbonyl compound. Carbonyl compound must contain at least one α-H

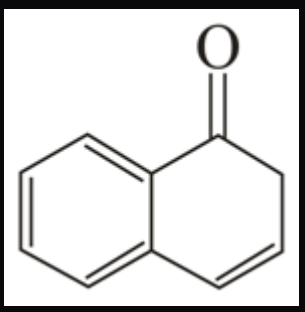
Compunds (i) , (v) have not alpha hydrogen so these compounds will not show enolization.

Compounds (ii),(iii),(iv) have alpha hydrogen so these compound wil show enolization.

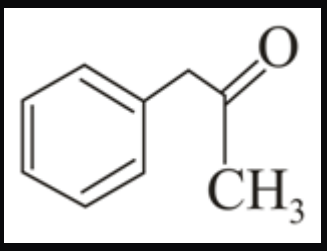
(i)



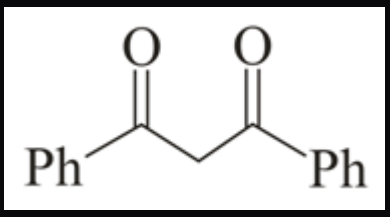
(ii)



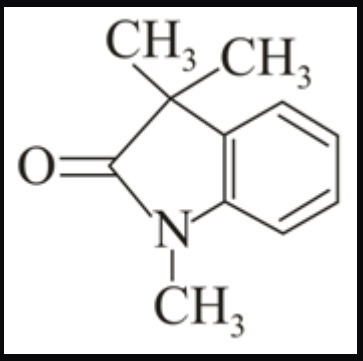
(iii)



(iv)



(v)



$(i) compound\ = 0\ (\alpha - H)$

$(ii) compound\ = 2\ (\alpha - H)$

$(iii) compound\ = 5\ (\alpha - H)$

$(iv) compound\ = 2\ (\alpha - H)$

$(v) compound\ = 0\ (\alpha - H)$

Q.44 Answer:

$3 : 4$

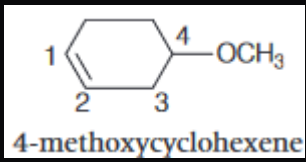
Solution:

Hint : $\frac{\lambda_1}{\lambda_2} = \frac{m_2\ \mathrm{V}_2}{m_1\ \mathrm{V}_1} = \frac{50 \times 1.5\ \mathrm{V}_1}{100 \times \mathrm{V}_1} = \frac{1.5}{2} = \frac{3}{4}$

Q.45 Answer:

4-methoxycyclohexene

Solution:



4-methoxycyclohexene

Note We give priority to double bond in numbering.

Q.46 Answer:

—

Solution:

$C_p = q/T;$

Heat required = No. of moles $\times C_p \times T$

$= (144 \times 25 \times 10)/(14.4 \times 2) = 1250\ \text{J}$

Q.47 Answer:

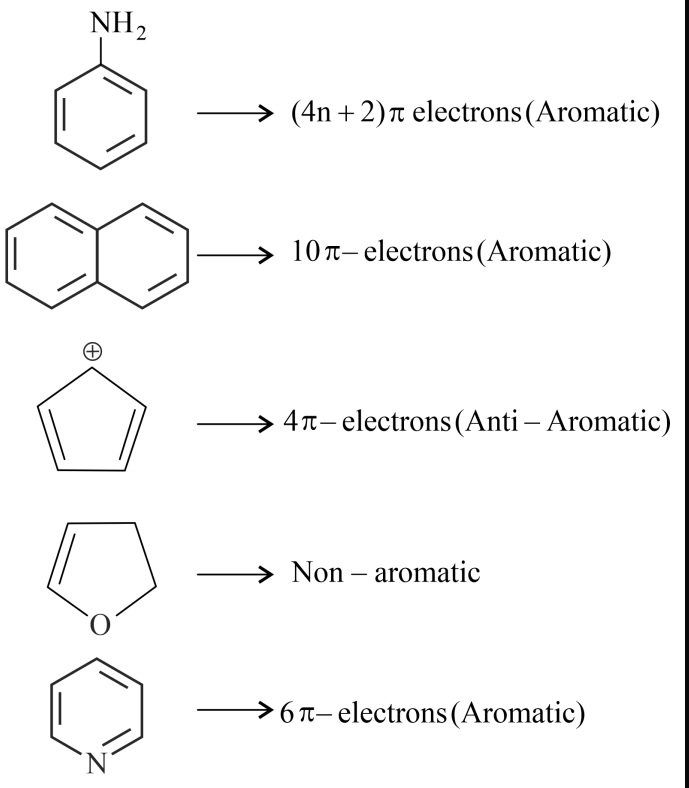
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Solution:

According to the Huckle's rule,

The compound is aromatic which is planar and has the $(4n + 2)\pi$ **electrons. The compound is aromatic in nature because it is a planar compound with** $(4n + 2)\pi$ **electrons. Aniline, naphthalene and pyridine are aromatic compounds.**

Compounds :-



Hence, the three compounds are aromatic in nature.

Q.48 Answer:

—

Solution:

$3\mathrm{d} \rightarrow$**radial nodes** $(\mathrm{n}-1-1)=0$

Angular nodes$=\mathrm{l}=2$

$4\mathrm{d} \rightarrow$**radial nodes**$=1$

Angular nodes $\mathrm{l}=2$

$5\mathrm{f} \rightarrow$**radial nodes**$=1$

Angular nodes$=3$

$3\mathrm{p} \rightarrow$**radial nodes**$=1$

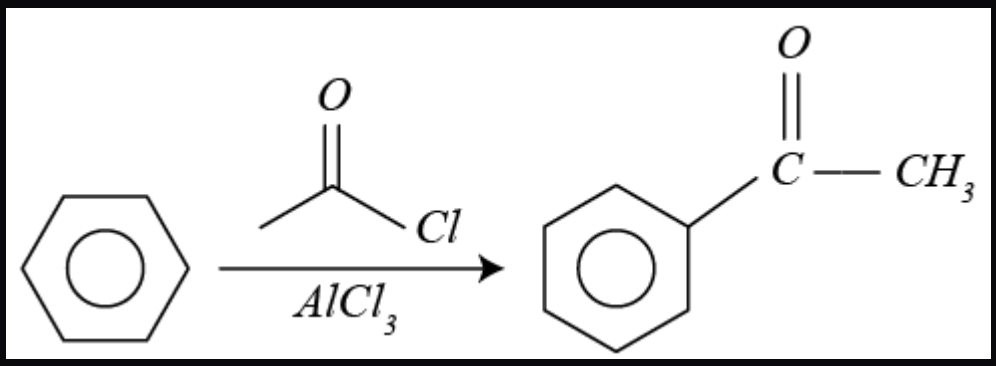
Angular nodes $=1$

Only $4\mathrm{d}$ **orbital has one radial node and** 2 **angular nodes.**

Q.49 Answer:

—

Solution:



$\frac{7.8}{78}$ mole $=0.1$ $\quad \frac{8.4}{120}$ mole $=0.07$

$\%$ yield $=\frac{0.07}{0.1} \times 100=20\%$

Q.50 Answer:

—

Solution:

From two data, (for zero order kinetics)

$\mathrm{K_I}=\frac{\mathrm{x}}{\mathrm{t}}=\frac{0.25}{0.05}=5$

$\Rightarrow K_{\mathrm{II}}=\frac{\mathrm{x}}{\mathrm{t}}=\frac{0.60}{0.12}=5$

Q.51 Answer:

$\frac{x^2}{2} \log x-\frac{x^2}{4}+C$

Solution:

$\because\ f(x)=\lim_{n\to\infty} n^2\left(\frac{1}{x^n}-\frac{1}{x^{n+1}}\right)$

Let $m=\frac{1}{n}$. So, as $n\to\infty \Rightarrow m\to 0$

$\Rightarrow f(x)=\lim_{m\to 0}\frac{1}{m^2}\left(x^m-x^{\frac{m}{m+1}}\right)$

$\Rightarrow \lim_{m\to 0}\frac{x^{\frac{m}{m+1}}}{m^2}\left[x^{\frac{m^2}{m+1}}-1\right]$

$\Rightarrow \lim_{m\to 0} x^{\frac{m}{m+1}}\left[\frac{x^{\frac{m2}{m+1}}-1}{\left(\frac{m^2}{m+1}\right)}\right]\cdot\frac{1}{(m+1)}=x^{\circ}[\log x]\cdot\frac{1}{(0+1)}$

$=1\cdot\log x\cdot 1\Rightarrow f(x)=\log x$

Now, $\int xf(x)dx=\int x\log x dx$

$=\log x\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot\frac{1}{x}dx+c=\frac{x^2}{2}\log x-\frac{x^2}{4}+c$

Q.52 Answer:

8

Solution:

We have,

$$f(x)=\begin{cases}1+x, & 0\le x\le 2\\ 3-x, & 2<x\le 3\end{cases}$$

$$f(f(x))=\begin{cases}1+f(x), & 0\le f(x)\le 2\\ 3-f(x), & 2<f(x)\le 3\end{cases}$$

$$ff(x)=\begin{cases}1+(1+x), & 0\le x\le 1\\ 3-(1+x), & 1<x\le 2\\ 1+(3-x), & 2<x\le 3\end{cases}$$

$$fof(x)=\begin{cases}2+x, & 0\le x\le 1\\ 2-x, & 1<x\le 2\\ 4-x, & 2<x\le 3\end{cases}$$

$\therefore fof(x)$ is discontinuous at $x=1$, and $x=2$

$\therefore\quad a=1, b=2$

$2a+3b=2(1)+3(2)=2+6=8$

Q.53 Answer:

7

Solution:

We have, $a+b+c=21$ and $\frac{a+c}{2}=b$

$\Rightarrow \quad a+c+\frac{a+c}{2}=21$

$\Rightarrow \quad a+c=14 \Rightarrow \frac{a+c}{2}=7$

$\Rightarrow \quad b=7$

So, a can take values from 1 to 6 . C can have values from 8 to 13

or a=b=c=7

So, there are 7 such triplets.

Q.54 Answer:

$a_2=-1$

Solution:

$a_1=a_3-2d$	$a_5=a_3+2d$
$a_3=-4$	$a_3\left({a_3}^2-4d^2\right)=80$
	$16-4d^2=-20$
	$d=\pm 3$
$d=3$	$(\because$ increasing A.P)

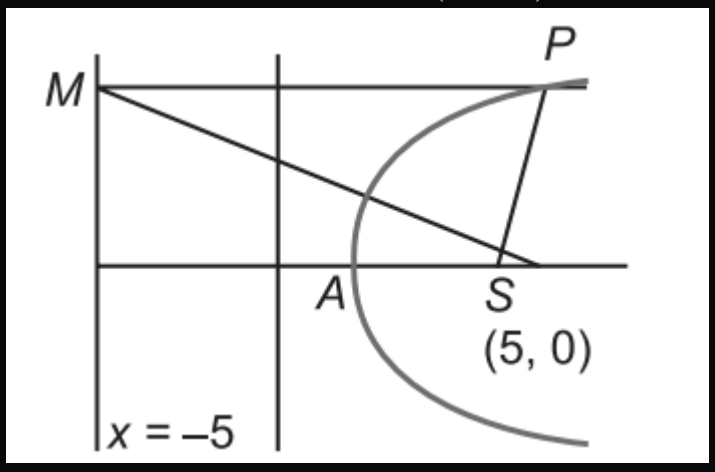
Q.55 Answer:

$(9,4\sqrt{3})$

Solution:

Given, that the $\triangle SPM$ (which is shown in figure) is equilateral.

Also, given parabola is $y^2=8(x-3)$ focus of this parabola is $S(5,0)$ and vertex $A(3,0)$.



Let coordinate of $P\left(h+at^2,k+2at\right)$

$=P\left(3+2t^2,4t\right)$

Then, coordinate of $M(-5,4t)$.

We know that the side of this equilateral triangle is $4a=4\times 2=8$

Now, $PS=8$

$\sqrt{\left(3+2t^2-5\right)^2+(4t)^2}=8$

$\Rightarrow \quad \sqrt{\left(2t^2-2\right)^2+(4t)^2}=8$

$\Rightarrow \quad \sqrt{(2t^2+2)^2} = 8$

$\Rightarrow \quad 2t^2 + 2 = 8$

$\Rightarrow \quad 2t^2 = 6$

$\Rightarrow \quad t = \sqrt{3}$

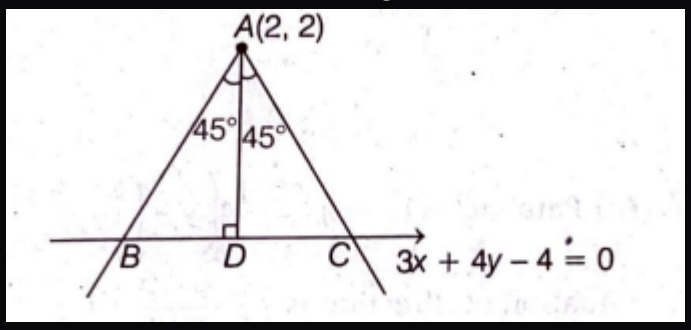
$\therefore P(3 + 2 \times 3, 4 \times \sqrt{3}) = P(9, 4\sqrt{3})$

Q.56 Answer:

$7x + y - 16 = 0$

Solution:

Equation of BC is $3x + 4y - 4 = 0$



$\therefore$ Slope of $AD = \frac{4}{3}$

Let the slope of one of other sides be m.

$$\text{Then, } \tan 45^\circ = \left|\frac{m - \frac{4}{3}}{1 + \frac{4m}{3}}\right| \Rightarrow 1 = \left|\frac{3m-4}{3+4m}\right|$$

$$\Rightarrow \quad \frac{3m-4}{3+4m} = 1 \text{ or } \frac{3m-4}{3+4m} = -1$$

$$\Rightarrow \quad m = -7 \text{ or } m = \frac{1}{7}$$

and side passes through $(2, 2)$.

$\therefore$ Equation of side can be

$7x + y - 16 = 0$ or $x - 7y + 12 = 0$

Q.57 Answer:

2

Solution:

$$\sum_{k=1}^{\infty}\sum_{r=0}^{k}\frac{1}{3^k}\left({}^kC_r\right)$$
$$=\sum_{k=1}^{\infty}\frac{1}{3^k}\left({}^kC_0+{}^kC_1+{}^kC_2+\ldots+{}^kC_k\right)$$
$$=\sum_{k=1}^{\infty}\frac{2^k}{3^k}=\sum_{k=1}^{\infty}\left(\frac{2}{3}\right)^k$$
$$=\frac{2}{3}+\left(\frac{2}{3}\right)^1+\ldots\infty$$
$$=\frac{2/3}{1-\frac{2}{3}}\quad\left(\because\text{ common ratio }=\frac{2}{3}\right)$$
$$=2$$

Q.58 Answer:

$2\left(1-e^{-1}\right)$

Solution:

Let $I=\int_{-1}^{1}\left\{\frac{x^{2013}}{e^{|x|}(x^2+\cos x)}+\frac{1}{e^{|x|}}\right\}dx$

$\Rightarrow \quad I=\int_{-1}^{1}\frac{x^{2013}}{e^{|x|}+\left(x^2+\cos x\right)}dx+\int_{-1}^{1}\frac{1}{e^{|x|}}dx$

Here, $\frac{x^{2013}}{e^{|x|}(x^2+\cos x)}$ is an odd function

and $\frac{1}{e^{|x|}}$ is an even function. $\left\{\because\int_{-a}^{a}f(x)dx=\begin{cases}2\int_0^9(x)dx; & f(x)\text{ is even}\\ 0, & f(x)\text{ is odd}\end{cases}\right\}$

$\therefore \quad 1=0+2\int_0^1 e^{-x}dx=-2(e^{-x})\Big\}_0^1=-2\left(e^{-1}\right)$

$=2\left(1-e^{-1}\right)$

Q.59 Answer:

3

Solution:

Given,

$\overrightarrow{a}=\hat{\mathrm{i}}+\hat{\mathrm{j}}+\widehat{\mathrm{k}}$ and $\overrightarrow{b}$ be two vectors such that $\overrightarrow{a}\cdot\overrightarrow{b}=1$ & $\cos\left(\overrightarrow{a}\cdot\overrightarrow{b}\right)=\frac{1}{3}$

So, applying formula of dot product we get,

$\overrightarrow{a}\cdot\overrightarrow{b}=1\Rightarrow\left|\overrightarrow{a}\right|\left|\overrightarrow{b}\right|\times\frac{1}{3}=1$

$\Rightarrow\sqrt{3}\left|\overrightarrow{b}\right|\times\frac{1}{3}=1$

$\Rightarrow\left|\overrightarrow{b}\right|=\sqrt{3}$

Also given that the components of $\overrightarrow{b}$ w.r.t $\left(\hat{\mathrm{i}},\ \hat{\mathrm{j}},\ \widehat{\mathrm{k}}\right)$ be integers,

Then the possible vectors that represent $\bar{b}$ will be $\left\{\left(\hat{\mathrm{i}}-\hat{\mathrm{j}}+\widehat{\mathrm{k}}\right),\ \left(-\hat{\mathrm{i}}+\hat{\mathrm{j}}+\widehat{\mathrm{k}}\right),\ \left(\hat{\mathrm{i}}+\hat{\mathrm{j}}-\widehat{\mathrm{k}}\right)\right\}$

Then the number of possible vectors that represent $\vec{b}$ is 3

Q.60 Answer:

one-one but not onto

Solution:

Given that, $A_n = \{(n+1)k \mid k \in \mathbf{N}\}$
If $n = 1, A_1 = \{2k \mid k \in \mathbf{N}\}$
for, $k = 1, 2, 3 \ldots A_1 = \{2, 4, 6, 8, \ldots\}$
If $n = 2, A_2 = \{3k \mid k \in \mathbf{N}\}$
for $k = 1, 2, 3, \ldots \quad A_2 = \{3, 6, 9, \ldots\}$
If $n = 3, A_3 = \{4k \mid k \in \mathbf{N}\}$
for $k = 1, 2, 3, \ldots A_3 = \{4, 8, 12, \ldots\}$
Now, $X = \bigcup_{n \in \mathbf{N}} A_n$ or $X = A_1 \cup A_2 \cup A_3 \ldots$
or $X = \{2, 3, 4, 5, 6, \ldots\}$
Here, $f : X \to N$ and $f(x) = x \forall x \in \mathbf{X}$
$\because f(x)$ is linear function i.e., it is one-one function but there is no value of x in domain where $f(x) = 1$
$\therefore f(x)$ is not onto function.

Q.61 Answer:

$\frac{4}{3}$

Solution:

Equation of line L_1 is
$\mathbf{r} = (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}) + \lambda(3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}})$
Equation of line L_2 is
$\mathbf{r} = (\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}) + \mu(\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 4\hat{\mathbf{k}})$
Distance between L_1 and L_2

$$D = \frac{(4\hat{\mathbf{j}}) \cdot [(3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) \times (\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 4\hat{\mathbf{k}})]}{|(3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) \times (\hat{\mathbf{i}} - \hat{\mathbf{j}}) - 4\hat{\mathbf{k}})|}$$

$$(3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) \times (\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 4\hat{\mathbf{k}}) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 2 & -2 \\ 1 & -6 & -4 \end{vmatrix}$$

$$= -(20\hat{\mathbf{i}} - 10\hat{\mathbf{j}} + 20\hat{\mathbf{k}})$$

$$D = \left| \frac{4\hat{\mathbf{j}} \cdot (-20\hat{\mathbf{i}} + 10\hat{\mathbf{j}} - 20\hat{\mathbf{k}})}{\sqrt{(20)^2 + (10)^2 + (20)^2}} \right| = \frac{4}{3}$$

Q.62 Answer:

$\sqrt{3} + \frac{1}{2}i$

Solution:

We have, $|z| \leq 2$

$$\Rightarrow \qquad \sqrt{x^2 + y^2} \leq 2 \quad \Rightarrow \quad x^2 + y^2 \leq 4 \qquad \ldots(\mathrm{i})$$

$\Rightarrow \quad (1-i)(x+iy)+(1+i)(x-iy) \geq 4$

$\Rightarrow \quad x+iy-ix+y+x-iy+ix+y \geq 4$

$\Rightarrow \quad 2x+2y \geq 4 \quad \Rightarrow \quad x+y \geq 2 \qquad ...(ii)$

So, $A \cap B = \{x^2+y^2 \leq 4\} \cap \{x+y \geq 2\}$

$\therefore \quad z = \sqrt{3}+\frac{1}{2}i$

Q.63 Answer:

5.5

Solution:

$$\bar{x} = \frac{\sum x_i}{5} \Rightarrow \sum x_i = 5 \times 2 = 10$$

$$\sigma_x^2 = \frac{\sum x_i^2}{5} - \bar{x}^2 \Rightarrow 4 + 2^2 = \frac{\sum x_i^2}{5} \Rightarrow \sum x_i^2 = 40$$

$$\bar{y} = \frac{\sum y_i}{5} \Rightarrow \sum y_i = 5 \times 4 = 20$$

$$\sigma_y^2 = \frac{\sum y_i^2}{5} - \bar{y}^2 \Rightarrow \frac{\sum y_i^2}{5} = 5 + 4^2 = 21 \Rightarrow \sum y_i^2 = 105$$

$$\bar{x} = \frac{\sum (x_i + y_i)}{10} = \frac{30}{10} = 3$$

$$\sigma^2 = \frac{\sum x_i^2 + y_i^2}{10} - 3^2 = \frac{145}{10} - 9 = 14.5 - 9 = 5.5$$

Q.64 Answer:

(4,5,-2)

Solution:

Equation of line joining points (0,-11,4) and (2,-3,1) is

$\frac{x-2}{2} = \frac{y+3}{8} = \frac{z-1}{-3} = \lambda$

Let P is any point of the above line then coordinate of P is $(2\lambda+2, 8\lambda-3, -3\lambda+1)$

$\therefore$ DR's **of PQ is** $(2\lambda+1, 8\lambda-11, -3\lambda-3$

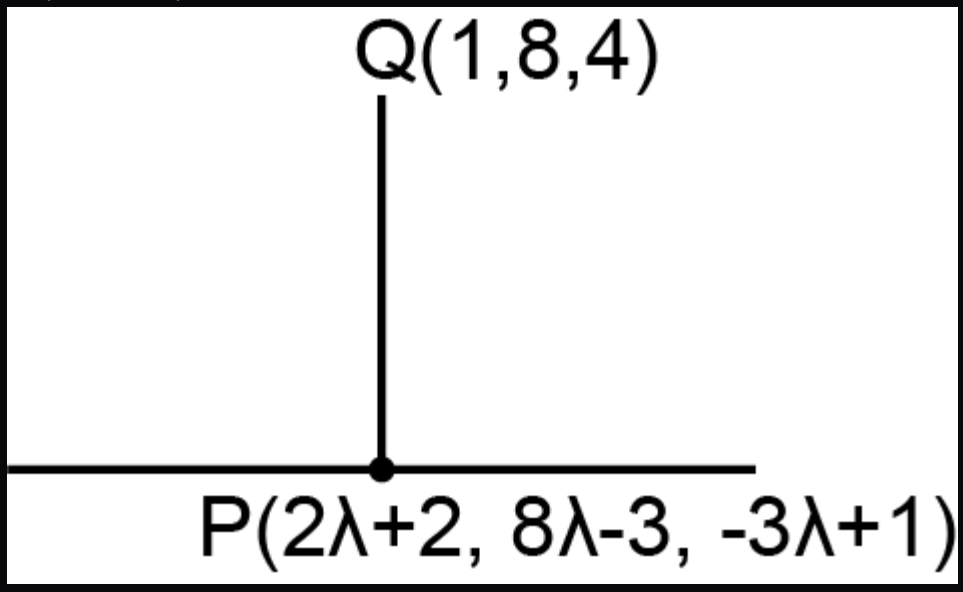
Now, $(2\lambda+1)(4+(8\lambda-11)(8)+(-3\lambda-3)(-3)=0$

$\Rightarrow \quad 4\lambda+2+64\lambda-88+9\lambda+9=0$

$77\lambda-77=0$

$\therefore$ **Required foot of perpendicular.**

$P(4, 5, -2)$



Q.65 Answer:

$\sqrt{1 - 2a + 2a^2}$

Solution:

Given, curve is

$2x^2 + y^2 = 2x$

$2x^2 - 2x + y^2 = 0$

$\Rightarrow \quad 2\left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{2}$

$\Rightarrow \quad \frac{\left(x - \frac{1}{2}\right)^2}{\frac{1}{4}} + \frac{y^2}{\frac{1}{2}} = 1$

which represents an ellipse.

Here, $a = \frac{1}{2}, b = \frac{1}{\sqrt{2}}, h = \frac{1}{2}, k = 0$

Consider a point $P(h + a\cos\theta, k + b\sin\theta)$

$= P\left(\frac{1}{2} + \frac{1}{2}\cos\theta, \frac{1}{\sqrt{2}}\sin\theta\right)$ **on the ellipse from which the distance of point** $(a, 0)$ **is maximum. let** $Q(a, 0)$

Now,

$PQ = \sqrt{\left(\frac{1}{2} + \frac{1}{2}\cos\theta - a\right)^2 + \left(\frac{1}{\sqrt{2}}\sin\theta - 0\right)^2}$

$= \sqrt{\frac{1}{4} + \frac{1}{4}\cos^2\theta + a^2 + \frac{1}{2}\cos\theta - a\cos\theta - a + \frac{1}{2}\sin^2\theta}$

$PQ = \sqrt{\frac{1}{2} + a^2 - a + \left(\frac{1}{2} - a\right)\cos\theta + \frac{1}{4}\sin^2\theta}$

$\Rightarrow$ **Let** $y = PQ^2$

$= \frac{1}{2} + a^2 - a + \left(\frac{1}{2} - a\right)\cos\theta + \frac{1}{4}\sin^2\theta$

For maxima and minima, put $\frac{dy}{d\theta} = 0$

$\Rightarrow -\left(\frac{1}{2} - a\right)\sin\theta + \frac{1}{4} \cdot 2\sin\theta\cos\theta = 0$

$\Rightarrow \quad \sin\theta\left(-\frac{1}{2} + a + \frac{1}{2}\cos\theta\right) = 0$

$\Rightarrow \sin\theta = 0$ **or** $-\frac{1}{2} + a + \frac{1}{2}\cos\theta = 0$

$\Rightarrow \quad \theta = 0$ **or** $\cos\theta = 1 - 2a$

$\Rightarrow \quad \sin^2\theta = 1 - \cos^2\theta$

$= 1 - (1 - 2a)^2$

$= -4a^2 + 4a$

Now, $\frac{d^2y}{d\theta^2} < 0$ **for** $\cos\theta = 1 - 2a$

Thus, distance PQ **is maximum, when** $\cos\theta = 1 - 2a$ **and** $\sin^2\theta = -4a^2 + 4a$

Now, required longest distance is

$= \sqrt{\frac{1}{2} + a^2 - a + \left(\frac{1}{2} - a\right)(1 - 2a) + \frac{1}{4}(-4a^2 + 4a)}$

$= \sqrt{\frac{1}{2} + a^2 - a + \frac{1}{2} - a - a + 2a^2 - a^2 + a}$

$= \sqrt{2a^2 - 2a + 1}$

$= \sqrt{1 - 2a + 2a^2}$

Q.66 Answer:

$y \sin y = x^2 \log x$

Solution:

$$\frac{dx}{dy} = \frac{\sin y(1 + y\cot y)}{x\log(\mathrm{x^2e})}, y(1) = 0$$

$$\Rightarrow \int x\log\left(x^2e\right)dx = \int (\sin y + y\cos y)dy$$

Let $x^2e = t$

$$\Rightarrow xdx = \frac{1}{2e}dt$$

$$\Rightarrow \frac{1}{2e}\int \log t dt = -\cos y + y\sin y - \int \frac{dy}{dy}\cdot \sin y dy + C$$

$$\Rightarrow \frac{1}{2e}[t\log t - t] = y\sin y + c$$

$$\Rightarrow \quad \frac{x^2e}{2e}\left(\log x^2e - 1\right) = y\sin y + c$$

$$\Rightarrow \quad \frac{x^2}{2}\left(\log x^2 + \log e - 1\right) = y\sin y + C$$

$$\Rightarrow \quad x^2\log x = y\sin y + C \quad \{\because \log_{\mathrm{e^e}} = 1\}$$

When $x = 1, y = 0$

$$\Rightarrow \quad 1\log 1 = 0 + C$$

$$\Rightarrow C = 0$$

Hence $x^2\log x = y\sin y$

Q.67 Answer:

'a' can attain only rational values

Solution:

Since, B **and** C **are the ends of diameter as** $\angle BAC$ **is** 90°

We know that equation of circle having endpoints of diameter as (x_1, y_1) **and** (x_2, y_2) **is given as**

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

Here, the endpoints of diameter are $(0,1)$ and $(1,0)$

$\therefore$ Equation of circle is

$x(x-1)+y(y-1)=0$

$\Rightarrow x^2+y^2-x-y=0$

Now, point D satisfies this equation

$\Rightarrow 4a^2+9a^2-5a=0$

$\Rightarrow a(13a-5)=0$

$\Rightarrow a=0,\ a=\frac{5}{13}$

Q.68 Answer:

$\frac{19}{3}$

Solution:

If $\mathrm{P}=\begin{bmatrix} a & b \\ c & d \end{bmatrix}=$ adj. $\mathrm{P}=\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

also $\mathrm{A}^{-1}=\frac{\text{adj}\cdot A}{\det\cdot A}$

$$\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1}=\frac{\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}}{2A^2+A-20}$$

$$\frac{1}{2A^2+A-20}\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}\begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix}=\begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$$

The gives, $\frac{A^2-5A-10}{2A^2+A-20}=14 \quad \mathrm{A}=3$ or $-\frac{10}{3}$

$3+\frac{10}{3}=\frac{19}{3}$

Q.69 Answer:

$\frac{1}{a}\left[\text{coefficient of } x^6 \text{ in } \left(ax^2+\frac{b}{x}\right)^{12}\right]$

Solution:

General term of $\left(a+\frac{b}{x^2}\right)$ is

$T_{r+1}={}^{11}C_r(ax)^{11-r}\times\left(\frac{b}{x^2}\right)^r$

$\Rightarrow T_{r+1}={}^{11}C_r\,a^{11-r}\cdot b^r(x)^{11-3r}$

For coefficient of x^{-7},

$11-3r=-7$

$\Rightarrow\ 3r=18$

$\Rightarrow\ \ r=6$

So,

$T_{6+1}={}^{11}C_6\,a^{11-6}\,b^6\ \ x^{-7}$

$T_{r+1}=\left({}^{11}C_6\,a^5b^6\right)x^{-7}$

So, $L={}^{11}C_6a^5b^6$

Similarly, general term of $\left(bx^2+\frac{a}{x}\right)^{11}$ is

$T_{r+1}={}^{11}C_r\left(bx^2\right)^{11-r}\times\left(\frac{a}{x}\right)^r$

$\Rightarrow T_{r+1}={}^{11}C_r\,b^{11-r}\times a^r\times(x)^{22-3r}$

So, for coefficient of x^7

$22 - 3r = 7 \Rightarrow r = 5$

Hence,

$M = {}^{11}C_5 b^6 a^5$

So, $L + M = 2 \times {}^{11}C_6 a^5 b^6 = 924 a^5 b^6$

And, general term of $\left(ax^2 + \frac{b}{x}\right)^{12}$ **is**

$T_{r+1} = {}^{12}C_r \left(ax^2\right)^{12-r} \left(\frac{b}{x}\right)^r$

$T_{r+1} = {}^{12}C_r \, a^{12-r} \, b^r \, x^{(24-3r)}$

For coeffiicent of x^6, $24 - 3r = 6 \Rightarrow r = 6$

So, coefficient of x^6 **is**

$= {}^{12}C_6 \, a^6 \, b^6 = 924 \, a^6 b^6$

Hence,

$L + M = \frac{1}{a} \times$ **[Coefficient of** x^6 **in** $\left(ax^2 + \frac{b}{x}\right)^{12}$**]**

Q.70 Answer:

0.75

Solution:

$$\mathrm{F(x)} + \mathrm{F}\left(\mathrm{x} + \frac{1}{2}\right) = 1$$

$$\mathrm{F}\left(\mathrm{x} + \frac{1}{2}\right) + \mathrm{F(x+1)} = 1$$

$$\mathrm{F(x)} - \mathrm{F(x+1)} = 0$$

$$\therefore \mathrm{F(x)} = \mathrm{F(x+1)}$$

$\mathrm{F(x)}$ **is periodic with period 1.**

$$\mathrm{I} = \int_0^{1500} \mathrm{F(x)} = 1500 \int_0^1 \mathrm{F(x)}$$

$$\Rightarrow 1500 \left[\int_0^{\frac{1}{2}} \mathrm{F(x)} + \int_{\frac{1}{2}}^1 \mathrm{F(x)dx}\right]$$

$$\Rightarrow 1500 \left[\int_0^{\frac{1}{2}} \mathrm{F(x)} + \int_0^{\frac{1}{2}} \mathrm{F}\left(\mathrm{x} + \frac{1}{2}\right)\right]$$

$$\Rightarrow 1500 \left[\int_0^{\frac{1}{2}} \left(\mathrm{F(x)} + \mathrm{F}\left(\mathrm{x} + \frac{1}{2}\right)\right) \mathrm{dx}\right]$$

$$1500 \left[\int_0^{\frac{1}{2}} \mathrm{dx}\right]$$

$$1500 \times \frac{1}{2} = 750$$

$$\therefore \mathrm{I} = \frac{750}{1000} = 0.75$$

Q.71 Answer:

—

Solution:

$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} - 3\hat{k}$

$A_1 =$ Area of the quadrilateral having $\vec{a}, \vec{b}$ as diagonals

$A_1 = \frac{1}{2}|\vec{a} \times \vec{b}|$

Now,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 1 & -2 & -3 \end{vmatrix}$$

$= \hat{i}(-6+6) - \hat{j}(-3-3) + \hat{k}(-2-2)$

$= 0\hat{i} + 6\hat{j} - 4\hat{k}$

$\Rightarrow \quad \vec{a} \times \vec{b} = 6\hat{j} - 4\hat{k}$

$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{36+16} = \sqrt{52}$

Then $A_1 = \frac{1}{2} \times \sqrt{52}$

Now, $A_2 =$ Area of parallelogram having $\vec{a}, \vec{b}$ as its adjacent sides

$\Rightarrow \quad A_2 = |\vec{a} \times \vec{b}| = \sqrt{52}$

So, $A_1 \cdot A_2 = \frac{1}{2} \times \sqrt{52} \times \sqrt{52} = \frac{52}{2} = 26.$

Q.72 Answer:

—

Solution:

Let events, E_1 : die A is used when head is appeared

E_2 : die B is used when tail is appeared

R : red face appears.

$\because \quad (P(E_1 \mid R) = \frac{32}{33}$

$\therefore \quad P(E_1) = \frac{1}{2}, P(E_2) = \frac{1}{2},$

$P(R \mid E_1) = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \ldots n \text{ times } = \left(\frac{2}{3}\right)^n$

Similarly, $P(R \mid E_2) = \left(\frac{1}{3}\right)^n$

$\because$ By Baye's theorem

$$P(E_1 \mid R) = \frac{P(E_1)\,P(R \mid E_1)}{P(E_1)\,P(R \mid E_1) + P(E_2)\,P(R \mid E_2)}$$

$$\Rightarrow \quad \frac{32}{33} = \frac{2^n}{1+2^n} \Rightarrow n = 5$$

Q.73 Answer:

—

Solution:

It is given that, if $x^2 + x + 1 = 0$, then

$A_r = \left(x + \frac{1}{x}\right)^3 \left(x^2 + \frac{1}{x^2}\right)^3 \left(x^3 + \frac{1}{x^3}\right)^3 \ldots \left(x^r + \frac{1}{x^r}\right)^3$

As, we know that w and w^2 are roots of quadratic equation $x^2 + x + 1 = 0$, so $w^3 = 1$ and

$1 + w + w^2 = 0$

$\therefore \quad A_r = \left(w + \frac{1}{w}\right)^3 \left(w^2 + \frac{1}{w^2}\right)^3 \left(w^3 + \frac{1}{w^3}\right)^3 \ldots \left(w^r + \frac{1}{w^r}\right)^3$

$\therefore \quad A_3 = (-1)(-1)(8) = 8$

Similarly, $A_6 = 8^2, A_9 = 8^3, \ldots$ **and so on.**

$\therefore \quad \frac{1}{A_3} + \frac{1}{A_6} + \frac{1}{A_9} + \frac{1}{A_{12}} + \ldots \infty$

$= \frac{1}{8} + \frac{1}{8^2} + \frac{1}{8^3} + \frac{1}{8^4} + \ldots \infty = \frac{\frac{1}{8}}{1 - \frac{1}{8}} = \frac{1}{7}$

Q.74 Answer:

—

Solution:

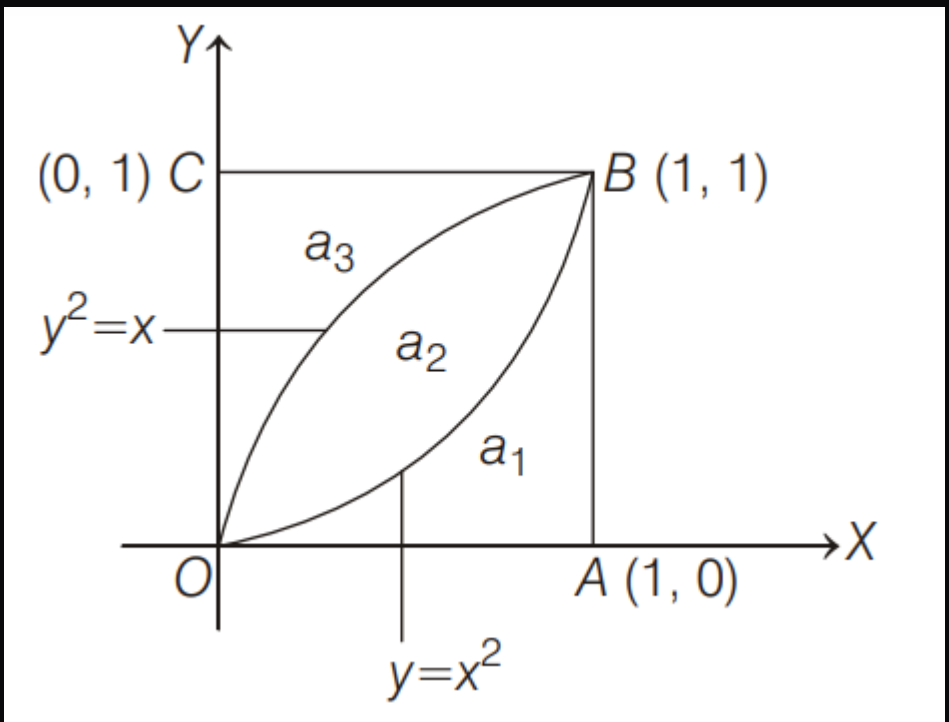
(b) $\because \qquad a_1 + a_2 + a_3 = 1 \qquad \ldots(\text{i})$

and due to symmetry $a_1 = a_3 \qquad \ldots(\text{ii})$

$$a_2 = \int_0^1 (\sqrt{x} - x^2)\, dx$$

Now,

$$= \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3}\right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$



So, $a_1 + a_3 + \frac{1}{3} = 1$

$\Rightarrow \qquad a_1 + a_3 = \frac{2}{3} \qquad \ldots(\text{ii})$

From Eqs. (ii) and (iii),

$a_1 = a_3 = \frac{1}{3} = a_2$

So, $a_1 + 2a_2 + 3a_3 = \frac{1}{3} + \frac{2}{3} + \frac{3}{3} = 2$

Q.75 Answer:

—

Solution:

$\text{Let } a + b + c + d = x \text{ and } e + f + g + h = y$

$x^2 + y^2 = (x + y)^2 - 2xy$

$= (13 + 6 + 4 + 2 - 2 - 3 - 5 - 7)^2 - 2xy$

$= 64 - 2xy$

As we know $\text{AM} \geq \text{GM}$

$$\frac{x + y}{2} \geq \sqrt{xy}$$

$$\Rightarrow \quad \frac{8}{2} \geq \sqrt{xy}$$

$$\Rightarrow \quad xy \leq 16$$

$x^2 + y^2$ is minimum when xy is maximum.

The maximum value of $xy = 16.$

The minimum value of $x^2 + y^2 = 64 - 2 \times 16 = 32$

$\therefore$ The minimum values of

$(a + b + c + d)^2 + (e + f + g + h) = 32$